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Quantum Signal Processing for Double-Quantum Control

Polynomials in the detuning, and the DQ π pulse on an unpolarised ^{14}N

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Outline

1. **Double quantum with an unpolarised ^{14}N.** The nucleus gives three detunings, and one π pulse cannot serve all three.
2. **Quantum signal processing.** Pulses and equal waits turn the detuning into a polynomial.
3. **QSP for the DQ π pulse.** The target is a step function of the detuning, and a polynomial approximates it. The same recipe prepares the ± 1 superposition and makes a universal NOT.

INTRO METHOD STRATEGY RESULTS ONGOING

01

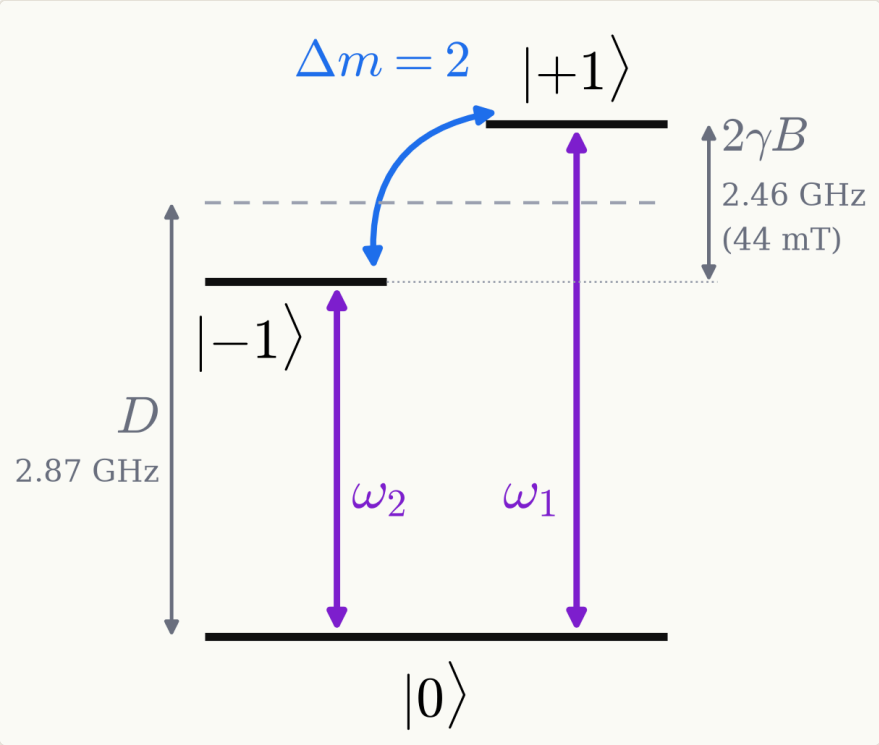
Double quantum with an unpolarised ^{14}N

Two frequency components, three hyperfine manifolds

INTRO

Double-quantum transition

Two frequency components drive the $\Delta m = 2$ transition between $|+1\rangle$ and $|-1\rangle$.



	superposition	phase after t
single quantum	$\|0\rangle + \|+1\rangle$	$\gamma B t$
double quantum	$\|-1\rangle + \|+1\rangle$	$2\gamma B t$

- One frequency component drives each transition $0 \leftrightarrow \pm 1$.
- The ± 1 superposition picks up twice the phase per unit field.
- Twice the phase doubles the magnetometric sensitivity.

Mamin et al., PRL 113, 030803 (2014)

INTRO

Rotating frame

The rotating frame leaves a differential and a common-mode detuning. In the laboratory frame, with a field B along the NV axis and one microwave field with two frequency components along x,

$$H(t) = DS_z^2 + \gamma B S_z + \left[\gamma B_1 \cos(\omega_1 t + \delta_1) + \gamma B_2 \cos(\omega_2 t + \delta_2)\right] S_x .$$

Rotate at ω_1 on $|+1\rangle$ and at ω_2 on $|-1\rangle$, and drop the terms that oscillate at frequencies of order D or $2\gamma B$ (rotating-wave approximation; Mamin et al., PRL 113, 030803 (2014)). In the basis $(|+1\rangle, |0\rangle, |-1\rangle)$,

$$\tilde{H} = \begin{pmatrix} \Delta_1 & \frac{\gamma B_1}{2\sqrt{2}} e^{-i\delta_1} & 0 \\ \frac{\gamma B_1}{2\sqrt{2}} e^{i\delta_1} & 0 & \frac{\gamma B_2}{2\sqrt{2}} e^{i\delta_2} \\ 0 & \frac{\gamma B_2}{2\sqrt{2}} e^{-i\delta_2} & \Delta_2 \end{pmatrix}, \qquad \Delta_{1,2} = D \pm \gamma B - \omega_{1,2}.$$

$$\mathrm{diag}(\Delta_1, 0, \Delta_2) = \Delta\, S_z + \Delta_c\, S_z^2, \qquad \Delta = \tfrac{\Delta_1 - \Delta_2}{2}, \quad \Delta_c = \tfrac{\Delta_1 + \Delta_2}{2}.$$

- The differential detuning Δ comes from the magnetic field and the hyperfine shift.
- The common-mode detuning Δ_c comes from the carrier offset from D, temperature and strain.

INTRO

Spin-1 representation

A balanced drive moves the spin-1 like a spin-½. Equal, mirrored components ($B_1 = B_2$, $\delta_1 = -\delta_2 = \varphi$, $\Delta_c = 0$, $\Omega_{DQ} = \gamma B_{\mathrm{mw}}/2$) give a spin-1 and a spin-½ in the same effective field:

$$\tilde{H} = \Delta S_z + \Omega_{DQ}(\cos\varphi\, S_x + \sin\varphi\, S_y) \quad \longleftrightarrow \quad h = \tfrac{\Delta}{2}\sigma_z + \tfrac{\Omega_{DQ}}{2}(\cos\varphi\,\sigma_x + \sin\varphi\,\sigma_y).$$

The three levels are the symmetric triplet of two spin-½, and $\mathbf{S} = \mathbf{s}_1 + \mathbf{s}_2$ puts the same field on both:

$$|{+1}\rangle = |{\uparrow\uparrow}\rangle, \quad |0\rangle = \tfrac{1}{\sqrt{2}}\big(|{\uparrow\downarrow}\rangle + |{\downarrow\uparrow}\rangle\big), \quad |{-1}\rangle = |{\downarrow\downarrow}\rangle, \qquad \tilde{H} = h \otimes \mathbb{1} + \mathbb{1} \otimes h \;\Rightarrow\; e^{-i\tilde{H}t} = e^{-iht} \otimes e^{-iht}.$$

Equivalently, S_i and $\sigma_i/2$ obey the same commutation relations, so one generator gives one rotation, as a 3×3 or a 2×2 matrix; classically, $\langle \mathbf{S} \rangle$ precesses like a Bloch vector. Hence

$$e^{-i\tilde{H}t} = D^1\big(e^{-iht}\big) \;\text{ for every } \Delta \quad \Longrightarrow \quad \text{a sequence of pulses and waits acts as } D^1(u).$$

INTRO

DQ π pulse

The DQ π pulse is $P = 0$ on the spin-½, and only Q^2 matters.

Spin-½: P = stay, Q = flip, $|P|^2 + |Q|^2 = 1$,

$$u = \begin{pmatrix} P & -Q^* \\ Q & P^* \end{pmatrix}.$$

Spin-1: the triplet block of $u \otimes u$,

$$D^1(u) = \begin{pmatrix} P^2 & -\sqrt{2}PQ^* & Q^{*2} \\ \sqrt{2}PQ & |P|^2 - |Q|^2 & -\sqrt{2}P^*Q^* \\ Q^2 & \sqrt{2}P^*Q & P^{*2} \end{pmatrix}.$$

DQ π pulse, $P = 0$:

$$D^1(u) = \begin{pmatrix} 0 & 0 & Q^{*2} \\ 0 & -1 & 0 \\ Q^2 & 0 & 0 \end{pmatrix}.$$

From $|0\rangle$, with $q = |Q|^2$:

$$P_0 = (1-2q)^2, \qquad P_{\pm 1} = 2q(1-q).$$

- Both spins flip: $|+1\rangle \to Q^2|-1\rangle$.
- $D^1(-u) = D^1(u)$, so only Q^2 matters.

INTRO

Hyperfine structure

The ^{14}N couples to the electron mainly through $A_{\parallel} S_z I_z$, which is block diagonal in m_I.

$$H = DS_z^2 + \gamma_e B S_z + A_{\parallel} S_z I_z + A_{\perp}(S_x I_x + S_y I_y) + P I_z^2 - \gamma_n B I_z + f(t)\left(S_x - \frac{\gamma_n}{\gamma_e} I_x\right)$$

term	size ($/2\pi$), at 44 mT	effect on the electron transitions
$A_{\parallel} S_z I_z$	-2.16 MHz	detuning $A_{\parallel} m_I$, comparable to the Rabi frequency
$A_{\perp}(S_x I_x + S_y I_y)$	-2.70 MHz	second-order shifts of 2 to 5 kHz
$P I_z^2$, $-\gamma_n B I_z$	-4.95 MHz, -135 kHz	same for every m_s: cancel
nuclear drive	7 kHz	far off resonance

Secular approximation: drop $A_{\perp}$ and the nuclear drive. Then H commutes with I_z:

$$H = \sum_{m=-1}^{1} H_m \otimes |m\rangle\langle m|, \qquad H_m = DS_z^2 + (\gamma_e B + A_{\parallel} m)\, S_z + f(t)\, S_x + \text{const.}$$

Doherty et al., Phys. Rep. 528, 1 (2013)

INTRO

Hyperfine manifolds

Each hyperfine manifold is a spin-1 with its own detuning. In the rotating frame, with the balanced drive, in the basis $(|+1\rangle, |0\rangle, |-1\rangle)$:

$$\tilde{H}_m = \begin{pmatrix} \Delta_m & \frac{\Omega_{DQ}}{\sqrt{2}} e^{-i\varphi} & 0 \\ \frac{\Omega_{DQ}}{\sqrt{2}} e^{i\varphi} & 0 & \frac{\Omega_{DQ}}{\sqrt{2}} e^{-i\varphi} \\ 0 & \frac{\Omega_{DQ}}{\sqrt{2}} e^{i\varphi} & -\Delta_m \end{pmatrix} = \Delta_m S_z + \Omega_{DQ}(\cos\varphi\, S_x + \sin\varphi\, S_y), \qquad \Delta_m = \Delta_0 + mA.$$

$$h_m = \tfrac{\Delta_m}{2}\sigma_z + \tfrac{\Omega_{DQ}}{2}(\cos\varphi\,\sigma_x + \sin\varphi\,\sigma_y), \qquad U = \sum_m D^1(u_m) \otimes |m\rangle\langle m|.$$

Unpolarised nucleus, $p_m = \frac{1}{3}$, $A/2\pi = 2.16$ MHz:

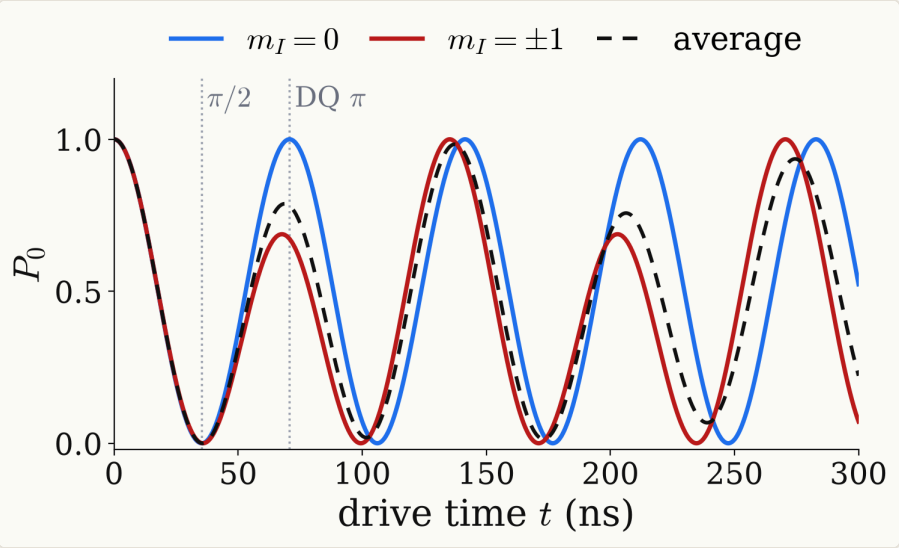
$$\rho_e \longmapsto \frac{1}{3} \sum_{m=-1}^{1} D^1(u_m)\, \rho_e\, D^1(u_m)^\dagger.$$

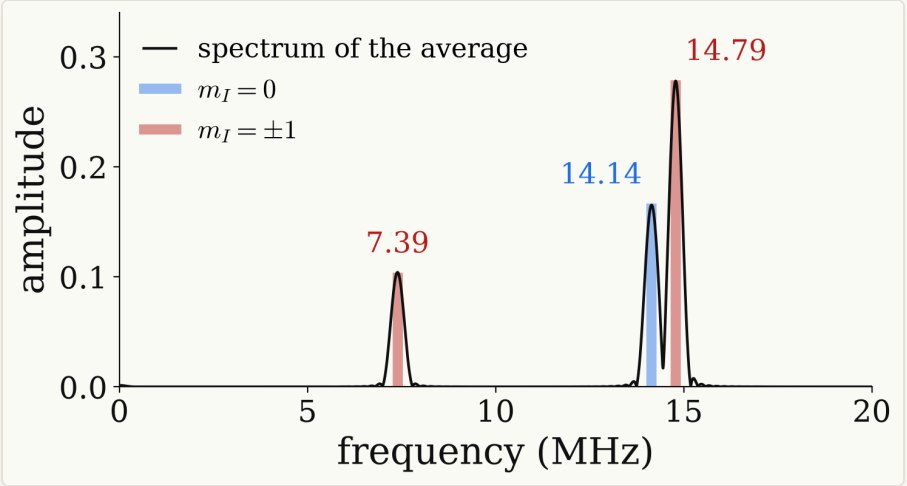
- The drive sits on the $m_I = 0$ resonance, $\Delta_0 = 0$; the outer manifolds see $\pm A$.
- No entanglement is created: the nucleus is a classical label.

INTRO

DQ nutation

The $|0\rangle$ population oscillates differently in each manifold: one line on resonance, two off resonance.





$$P_0^{(m)}(t) = \left[\frac{\Delta_m^2}{\Omega_m'^2} + \frac{\Omega_{DQ}^2}{\Omega_m'^2}\cos\Omega_m' t\right]^2, \quad \Omega_m' = \sqrt{\Omega_{DQ}^2 + \Delta_m^2}$$

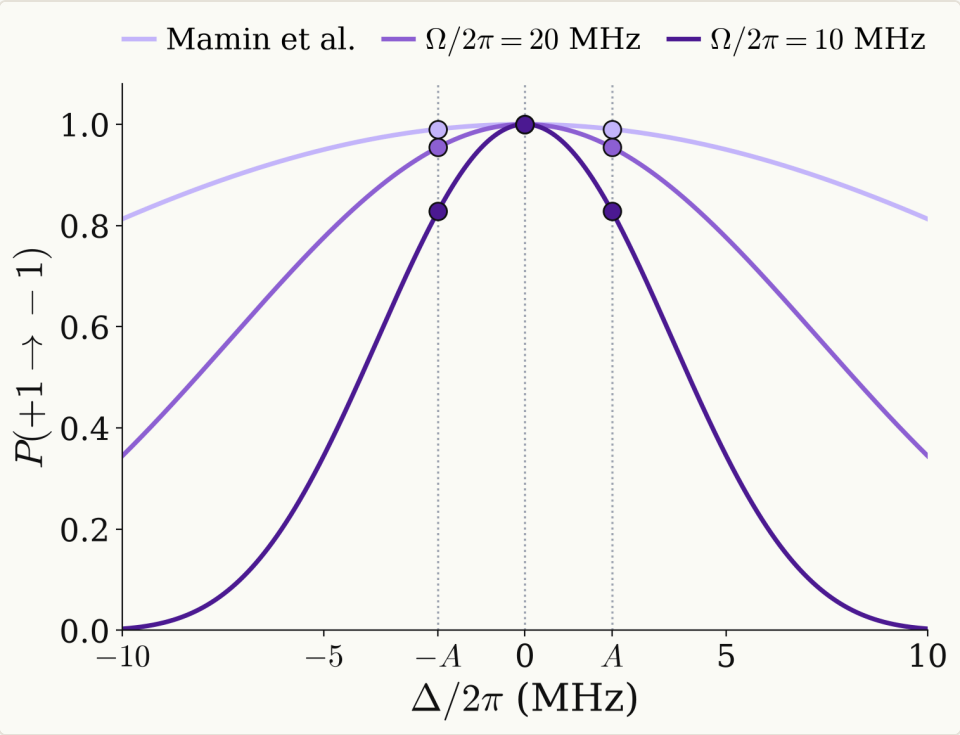
$$P_0^{(m)} = c_m + \frac{2\Delta_m^2\Omega_{DQ}^2}{\Omega_m'^4}\cos\Omega_m' t + \frac{\Omega_{DQ}^4}{2\Omega_m'^4}\cos 2\Omega_m' t$$

- On resonance $P_0 = \cos^2\Omega_{DQ} t$: a single line at $2\Omega_{DQ}$.
- Off resonance $q < 1$: P_0 returns only to 0.69 at the DQ π time, and a line at Ω_m' appears.

INTRO

Single-pulse error

One rectangular DQ π pulse misses the outer hyperfine manifolds.



$$q = 1 - \eta_m^2 + O(\eta^4), \qquad 1 - P_{-1} \approx \tfrac{4}{3}\,\eta_A^2$$

with $\eta_m = \Delta_m/\Omega_{DQ}$ and $\Omega = \sqrt{2}\,\Omega_{DQ}$.

drive	η_A	$1 - P_{-1}$	$\pi/2$ phase
Mamin et al.	0.069	6.3×10^{-3}	$\pm 7.9^\circ$
$\Omega/2\pi = 20$ MHz	0.153	3.0×10^{-2}	$\pm 17.5^\circ$
$\Omega/2\pi = 10$ MHz	0.305	0.115	$\pm 34.9^\circ$

- The π pulse leaves population in $|0\rangle$.
- The $\pi/2$ pulse fails through its phase.

Phase coherence needs one common Q^2: $Q(\Delta_m) = s_m\, e^{i\chi}$ with signs $s_m = \pm 1$ and one phase χ.

02

Quantum signal processing

The waits fix the degree; the pulses choose the polynomial

METHOD

Free evolution

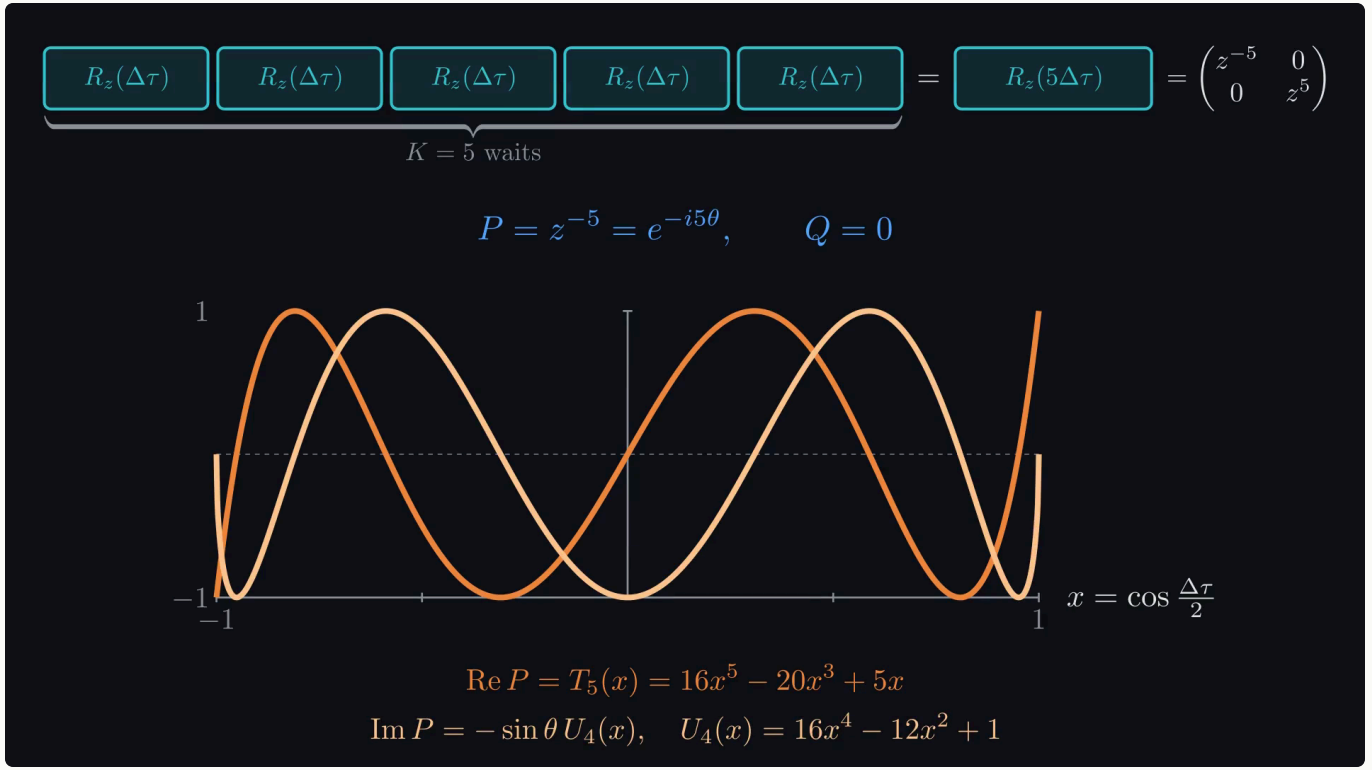
Without pulses, K waits make one rotation $R_z(K\Delta\tau)$.

$$\underbrace{R_z(\Delta\tau)\cdots R_z(\Delta\tau)}_{K \text{ waits}} = R_z(K\Delta\tau) = \begin{pmatrix} z^{-K} & 0 \\ 0 & z^{K} \end{pmatrix}$$

$$P = z^{-K} = e^{-iK\theta}, \qquad Q = 0$$

$$\operatorname{Re} P = \cos K\theta = T_K(x)$$
$$\operatorname{Im} P = -\sin K\theta = -\sin\theta\, U_{K-1}(x)$$

- $\theta = \frac{\Delta\tau}{2}$, $z = e^{i\theta}$, $x = \cos\theta$.
- T_K and U_{K-1} are polynomials of degree K and $K-1$ in x.



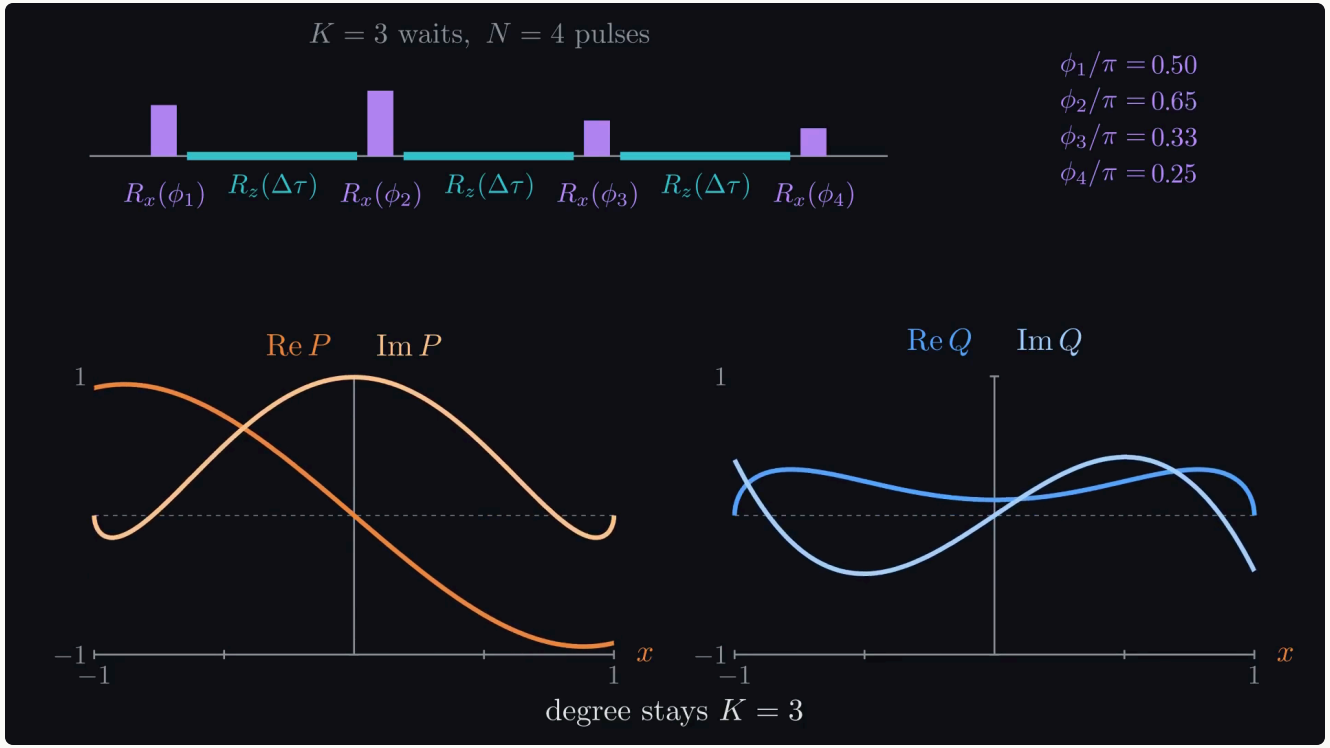
METHOD

Pulse sequence

Pulses between the waits tune P and Q; the degree stays K.

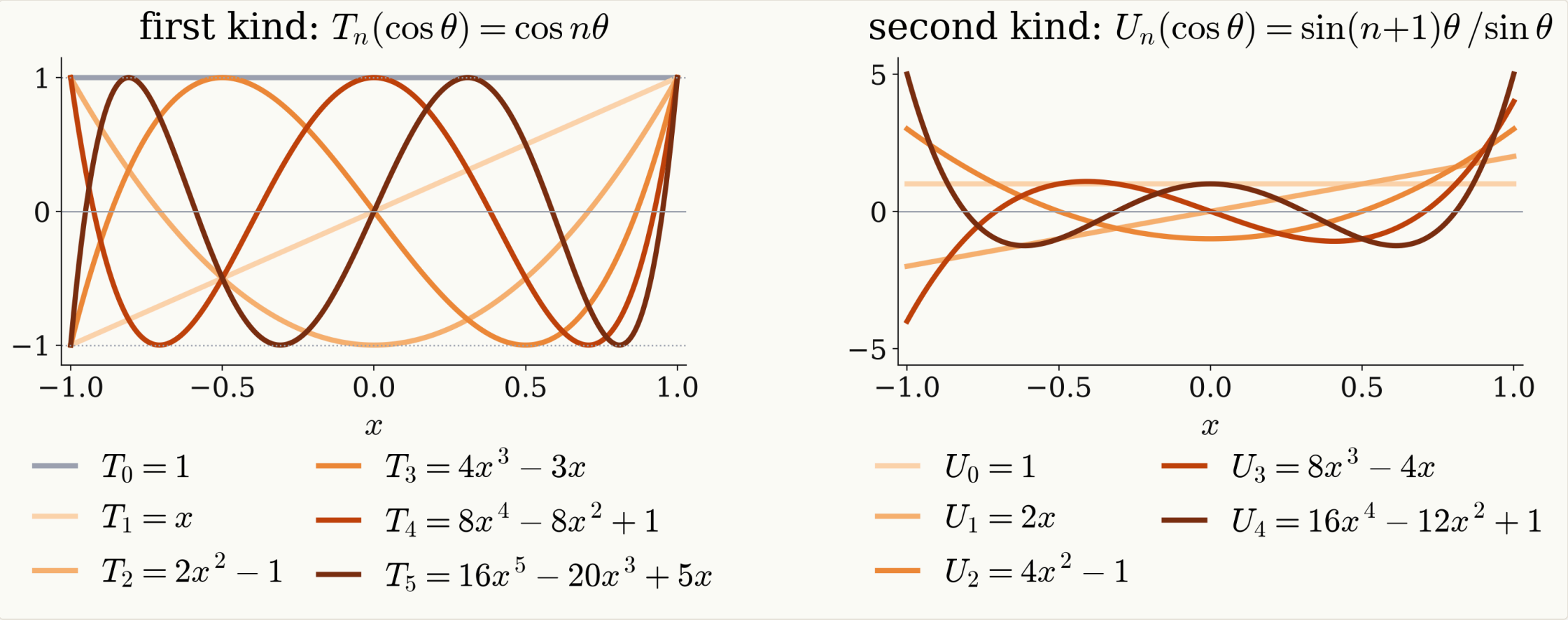
$$u = R_x(\phi_N)\, R_z(\Delta\tau)\, R_x(\phi_{N-1}) \cdots R_z(\Delta\tau)\, R_x(\phi_1) = \begin{pmatrix} P & -Q^* \\ Q & P^* \end{pmatrix}$$

- K waits and $N = K + 1$ pulses about x.
- Each angle ϕ_k changes the polynomials, not their degree.
- All pulses off gives back $\operatorname{Re} P = T_K(x)$ and $Q = 0$.



METHOD

Chebyshev polynomials

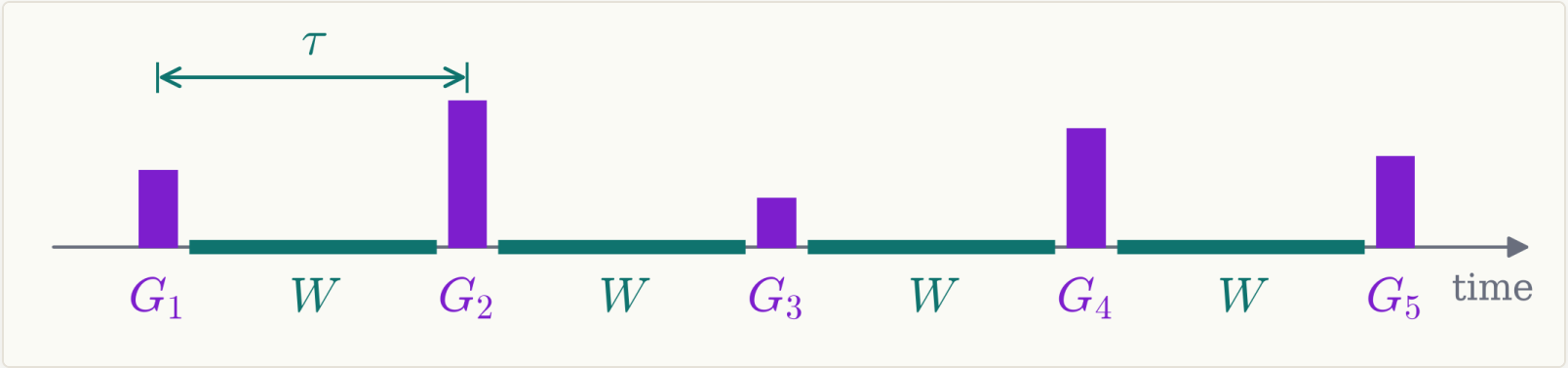


$$T_{n+1} = 2x\,T_n - T_{n-1} \quad \text{(same for } U\text{)} \quad \Longleftarrow \quad \cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos n\theta$$

METHOD

Signal and processing

The waits carry the signal; the pulses process it.



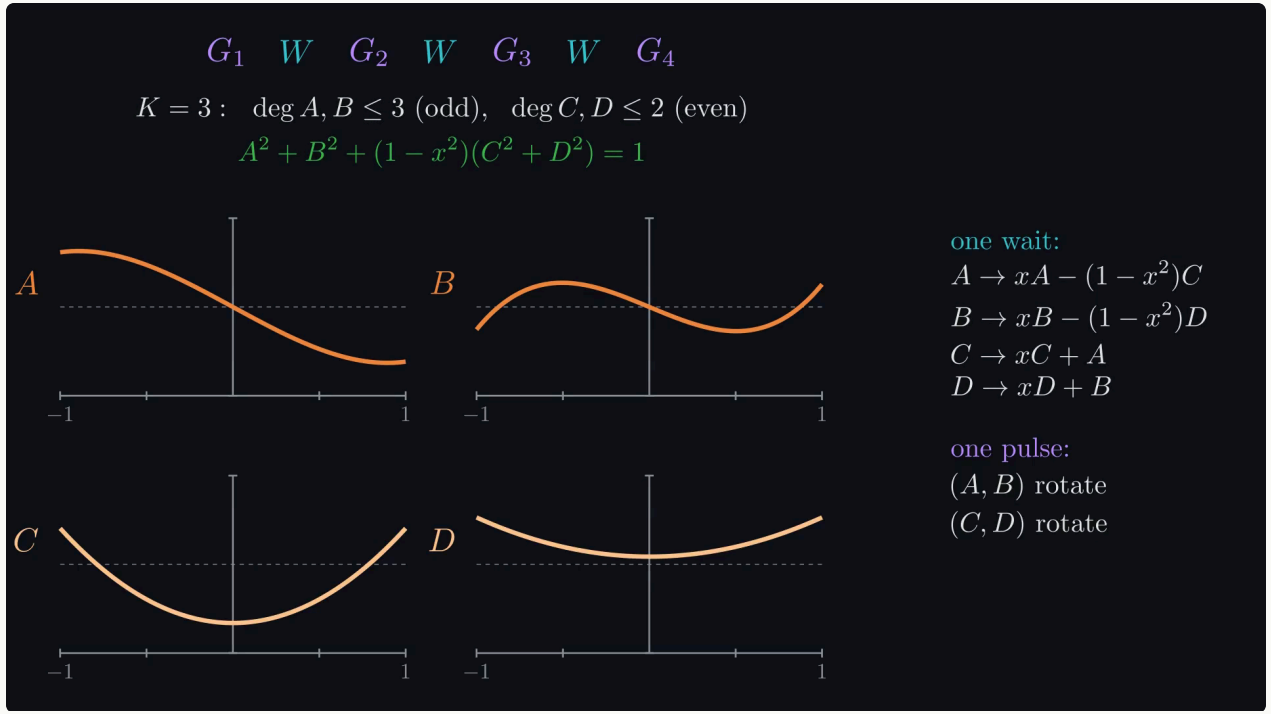
$$u = G_N\, W\, G_{N-1} \cdots W\, G_1 = \begin{pmatrix} P & -Q^* \\ Q & P^* \end{pmatrix}, \qquad W = R_z(\Delta\tau) = \begin{pmatrix} z^{-1} & 0 \\ 0 & z \end{pmatrix}, \qquad z = e^{i\theta},\ \theta = \frac{\Delta\tau}{2}$$

- Only the waits depend on Δ: each multiplies an amplitude by z^{-1} or z.
- The pulses G_k are instantaneous rotations about any axis in the xy plane.
- P and Q have degree $K = N - 1$, the number of waits.

METHOD Reachable polynomials

K waits and $K+1$ pulses about x make exactly these polynomials:

$$u = A\,\mathbb{I} - iB\,X - i\sin\theta\,(C\,Z + D\,Y), \qquad W = x\,\mathbb{I} - i\sin\theta\,Z, \qquad R_x(\phi) = \cos\tfrac{\phi}{2}\,\mathbb{I} - i\sin\tfrac{\phi}{2}\,X$$



real polynomials	degree	parity
$A,\ B$	$\le K$	of K
$C,\ D$	$\le K-1$	of $K-1$

$$A^2 + B^2 + (1-x^2)(C^2+D^2) = 1$$

$$P = A - i\sin\theta\,C, \quad Q = -iB + \sin\theta\,D$$

Conversely, every such A, B, C, D is made by $K+1$ pulses about x:

$$\underbrace{2(K+1)}_{\text{coefficients}} - \underbrace{(K+1)}_{\text{unitarity}} = \underbrace{K+1}_{\text{angles}}$$

Low, Yoder, Chuang, PRX 6, 041067 (2016); Martyn, Rossi, Tan, Chuang, PRX Quantum 2, 040203 (2021), Thm. 1

03

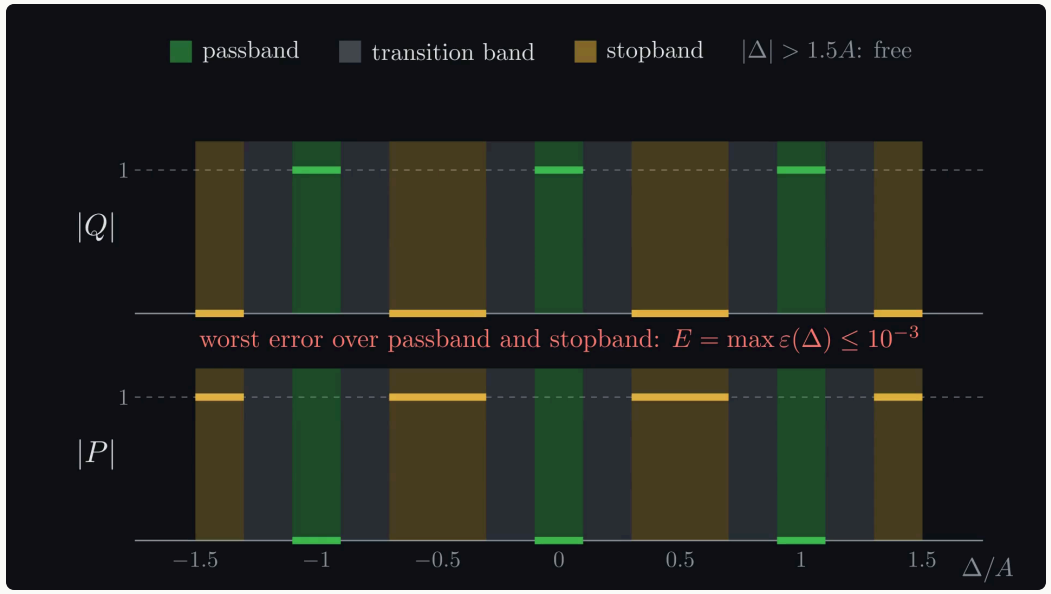
QSP for the DQ π pulse

Setting the target, choosing the wait, approximating a step

STRATEGY

Target function

The target is a step function of the detuning.



- **Passband** $|\Delta - mA| \le 0.1A$: invert, $P = 0$, $Q = s_m e^{i\chi}$ with one common χ.
- **Stopband** $0.3A \le |\Delta| \le 0.7A$ and $1.3A \le |\Delta| \le 1.5A$: nothing happens, $Q = 0$.
- **Transition bands** in between are free.

Error at one detuning, the process infidelity of the spin-½ operation:

$$\varepsilon(\Delta) = 1 - \left|\tfrac{1}{2}\mathrm{Tr}\, u_t^\dagger u\right|^2 = \begin{cases} |P|^2 + \left[\mathrm{Im}\left(e^{-i\chi}Q\right)\right]^2 & \text{passband: population left behind + phase error} \\ |Q|^2 & \text{stopband: population transferred} \end{cases}$$

Uniform error $E = \max_{\Delta \,\in\, \text{passband} \cup \text{stopband}} \varepsilon(\Delta)$, the worst case; we ask for $E \le 10^{-3}$.

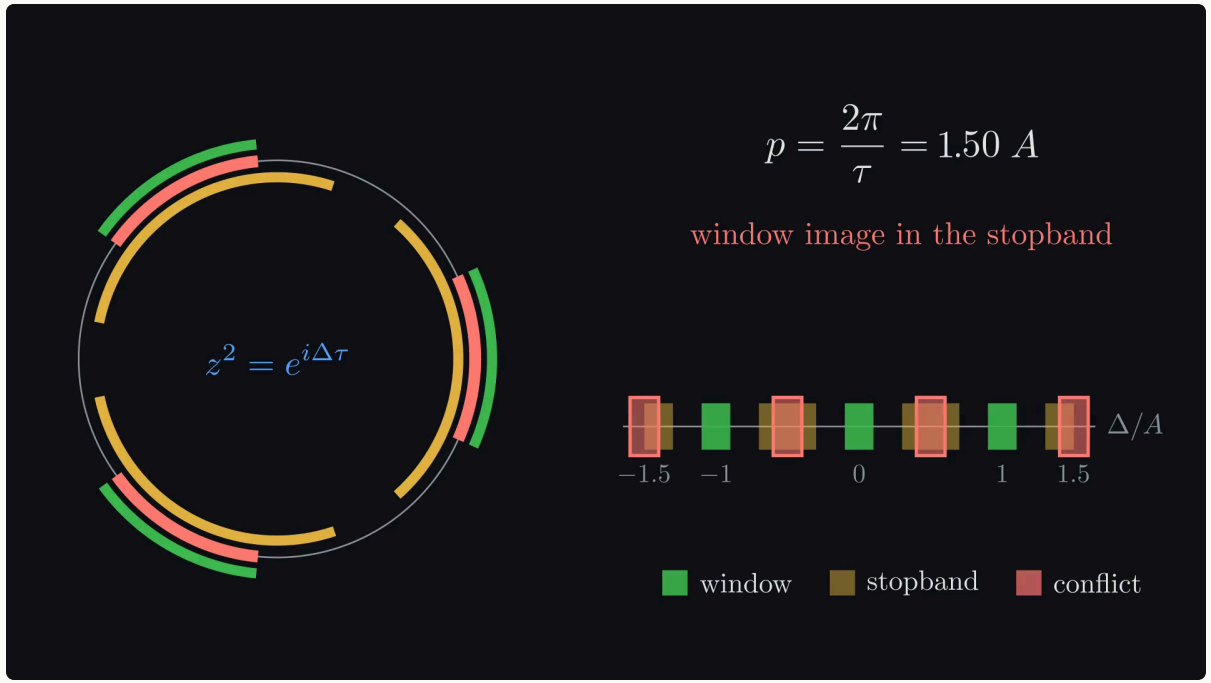
STRATEGY

Periodicity

The wait τ follows from the periodicity of the response. A wait depends on Δ only through $e^{\pm i\Delta\tau/2}$. Shifting Δ by $2\pi/\tau$ flips the sign of every wait:

$$\Delta \to \Delta + \tfrac{2\pi}{\tau}: \quad W \to -W, \quad u \to (-1)^K u, \quad Q^2 \to Q^2.$$

- The response repeats every $p = 2\pi/\tau$, and so do the windows.
- An image of a window must not land in the stopband.
- Compatible iff $0.9A < p < 1.1A$, $1.8A < p < 2.2A$ or $p > 2.6A$.



RESULTS

Choice of τ

The wait $\tau = 2\pi/A$ needs the fewest pulses.

τ	τ (ns)	period of Q^2	smallest N with $E \leq 10^{-3}$	free evolution $K\tau = (N-1)\tau$
$2\pi/A$	463.0	A	**14**	6019 ns
π/A	231.5	$2A$	27	6019 ns
$2\pi/3A$	154.3	$3A$	40	6019 ns
$2\pi/4A$	115.7	$4A$	51	5787 ns
$2\pi/6A$	77.2	$6A$	75	5710 ns
$2\pi/1.5A$	308.6	$1.5A$	$E \geq 0.50$ for any N	
$2\pi/2.5A$	185.2	$2.5A$	$E \geq 0.50$ for any N	

- At π/A and $2\pi/3A$ the optimum is the $2\pi/A$ sequence with zero-angle pulses in between.
- Shorter waits save 4 to 5 % of free evolution at four to five times the pulses.
- Incompatible τ ask one detuning to be inverted and unchanged: $E \geq 1/2$.

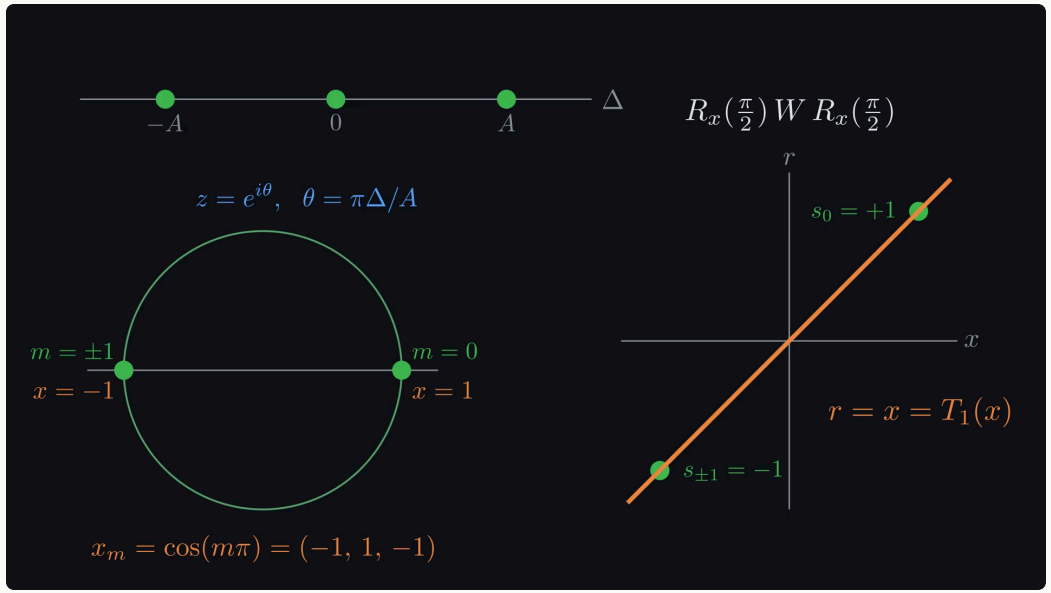
METHOD

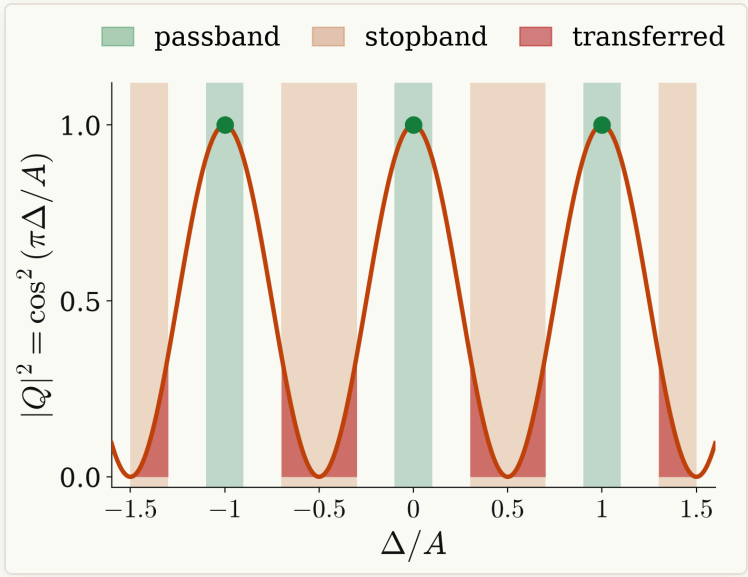
Simple targets

Two simple targets need only one or two pulses.

- **Passband only:** $r \equiv 1 = T_0$, $K = 0$. One $R_x(\pi)$ gives $P = 0$, $Q = -i$ at every Δ.
- **Exact at the three resonances**, $\tau = 2\pi/A$: $x_m = \cos m\pi = (-1, 1, -1)$, so $r = x = T_1$, $K = 1$:

$$R_x(\tfrac{\pi}{2})\, W\, R_x(\tfrac{\pi}{2}): \qquad P = -i \sin \tfrac{\Delta\tau}{2} = 0, \quad Q = -i \cos \tfrac{\Delta\tau}{2} = \mp i \quad \text{at every } \Delta = nA.$$





- Neither design protects the stopband: the second transfers up to 0.35 of the population there.

METHOD

Chebyshev expansion

The target is a Chebyshev series in x. Each wait contributes $z^{\pm 1}$, so $Q = \sum_k q_k z^k$ is a Laurent polynomial. Its harmonics pair into Chebyshev polynomials:

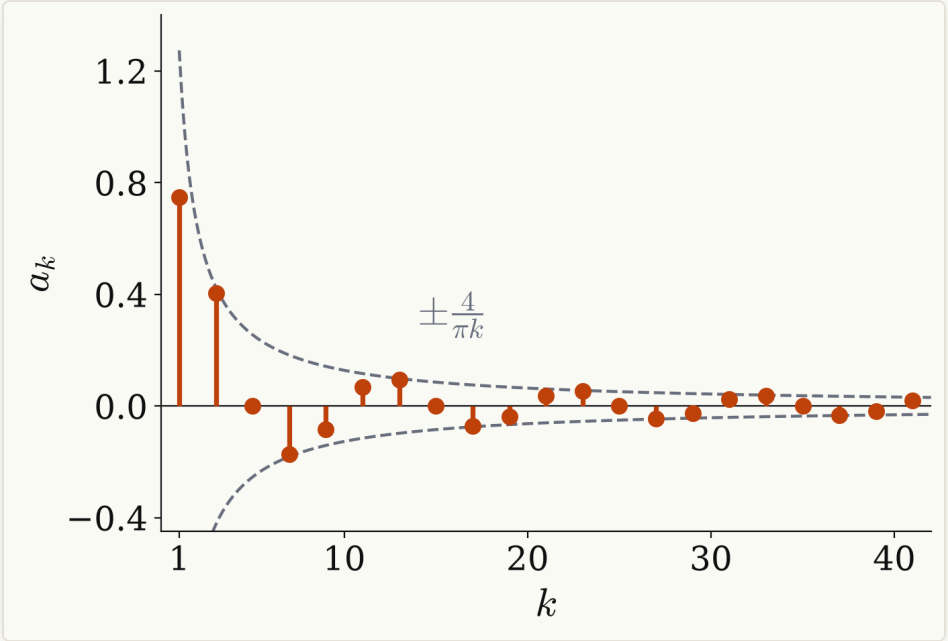
$$z^{\pm k} = T_k(x) \pm i \sin\theta \, U_{k-1}(x), \qquad x = \cos\theta.$$

The target $Q_t = -i\, r(\theta)$ is even in Δ, so $q_k = q_{-k}$ and only cosines survive:

$$r(\theta) = \sum_k a_k \cos k\theta = \sum_k a_k \, T_k(x).$$

At $\tau = 2\pi/A$ the step with its jumps mid-band has

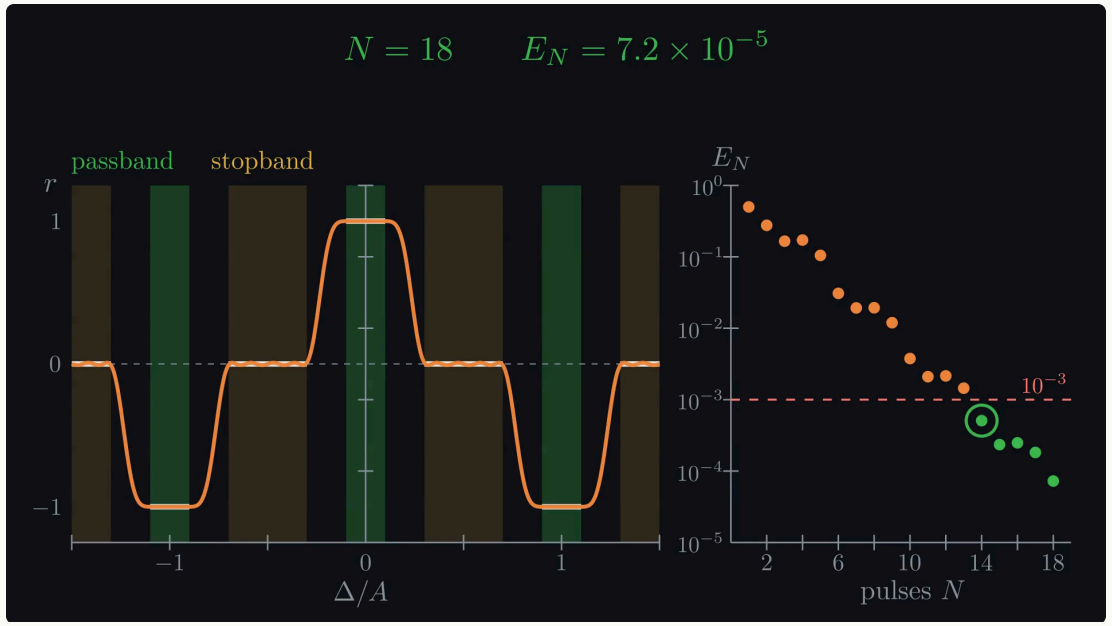
$$a_k = \frac{4 \sin(0.2\pi k)}{\pi k}, \qquad k \text{ odd}.$$



RESULTS

Best uniform approximation

The best polynomial gives the best achievable error E_N. Here $Q = -i\,r$ with one real series $r(x) = \sum_k c_k\,T_k(x)$ of degree K, made by $N = K + 1$ pulses.



condition	meaning
$s_m\,r \geq \sqrt{1-E}$ in the passband	at most E of the population is left behind
$\lvert r\rvert \leq \sqrt{E}$ in the stopband	at most E is transferred
$\lvert r\rvert \leq 1$ everywhere	all that unitarity asks; P follows

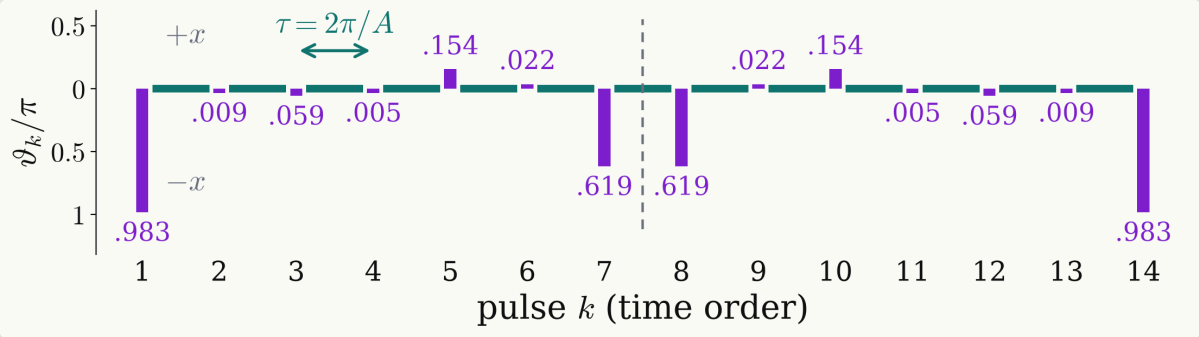
- For a fixed E the conditions are linear in the c_k: a linear program finds the smallest error E_N.
- $N = 14$ ($K = 13$) gives $E_{14} = 5.1 \times 10^{-4}$ in $6.0\,\mu\text{s}$.

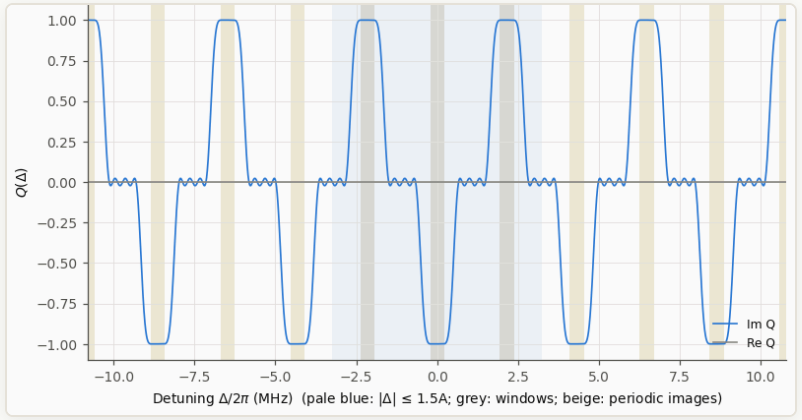
RESULTS

Pulse sequence ($N = 14$)

The best polynomial gives fourteen pulses that invert at every $\Delta = nA$:

$$r \xrightarrow{|P|^2 = 1 - r^2} (P, Q) \xrightarrow{\text{strip pulse and wait}} \vartheta_1, \ldots, \vartheta_{14}.$$





$$Q^2(\Delta + A) = Q^2(\Delta)$$

- The sequence is palindromic, about $\pm x$.
- It inverts at every $\Delta = nA$ and leaves the state unchanged in between.
- Free evolution $K\tau = 6019$ ns, $E = 5.2 \times 10^{-4}$.

METHOD

Symmetric QSP

A palindromic sequence leaves one real function.

$$u = G_1\, W\, G_2 \cdots G_2\, W\, G_1, \qquad W^T = W, \quad R_x(\vartheta)^T = R_x(\vartheta) \quad \Rightarrow \quad u^T = u$$

$$\begin{pmatrix} P & -Q^* \\ Q & P^* \end{pmatrix} = \begin{pmatrix} P & Q \\ -Q^* & P^* \end{pmatrix} \Rightarrow\ Q = -Q^* = -i\, r, \qquad r(x) = \sum_{k \equiv K\ (2)} c_k\, T_k(x)\ \ \text{real}$$

Symmetric QSP: the phase list is a palindrome, $(\phi_0, \phi_1, \ldots, \phi_1, \phi_0)$. It was proposed for computing phase factors by Dong, Meng, Whaley and Lin (2021); Wang, Dong and Lin (2022) proved that every admissible polynomial pair has exactly one symmetric phase list, and that the cost is strongly convex near a fixed symmetric starting point when the target is small.

- Every $Q(\Delta_m)$ has the phase $\pm\pi/2$, so phase coherence is automatic.
- The target becomes three conditions on one real function r.
- With instantaneous pulses, symmetric and arbitrary sequences reach the same error.

RESULTS

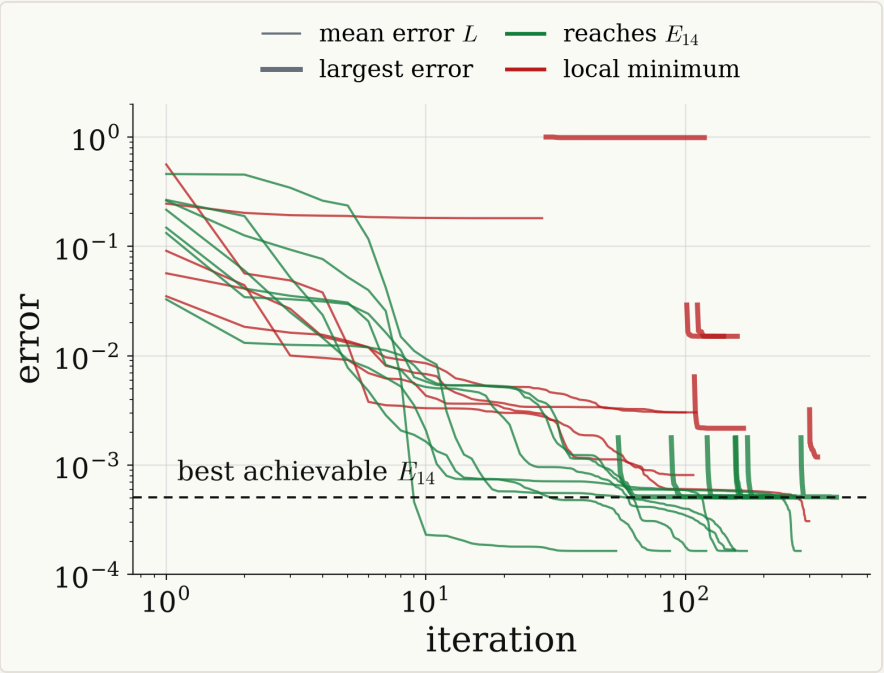
Loss function

Training on the mean error, then on the largest error, reaches the best achievable error.

$$L = \frac{1}{M}\Big[\underbrace{M_{\text{pass}} - \frac{1}{2}\Big(\sum_{j\in\text{pass}} |Q_j|^2 + \Big|\sum_{j\in\text{pass}} Q_j^2\Big|\Big)}_{\min_\chi \sum_{\text{pass}} \varepsilon_j} + \underbrace{\sum_{j\in\text{stop}} |Q_j|^2}_{\text{population transferred}}\Big]$$

$$\varepsilon_{\text{pass}} = \underbrace{|P|^2}_{\text{not transferred}} + \underbrace{\big[\text{Im}(e^{-i\chi}Q)\big]^2}_{\text{phase error}}$$

- The common phase χ is optimised in closed form, so only Q_j^2 enters.
- Stage 1 minimises the mean L (exact gradient); stage 2 the largest error.
- At $N = 14$, 7 of 12 random starts reach E_{14}; symmetric and arbitrary sequences reach E_N for every $N \geq 3$.

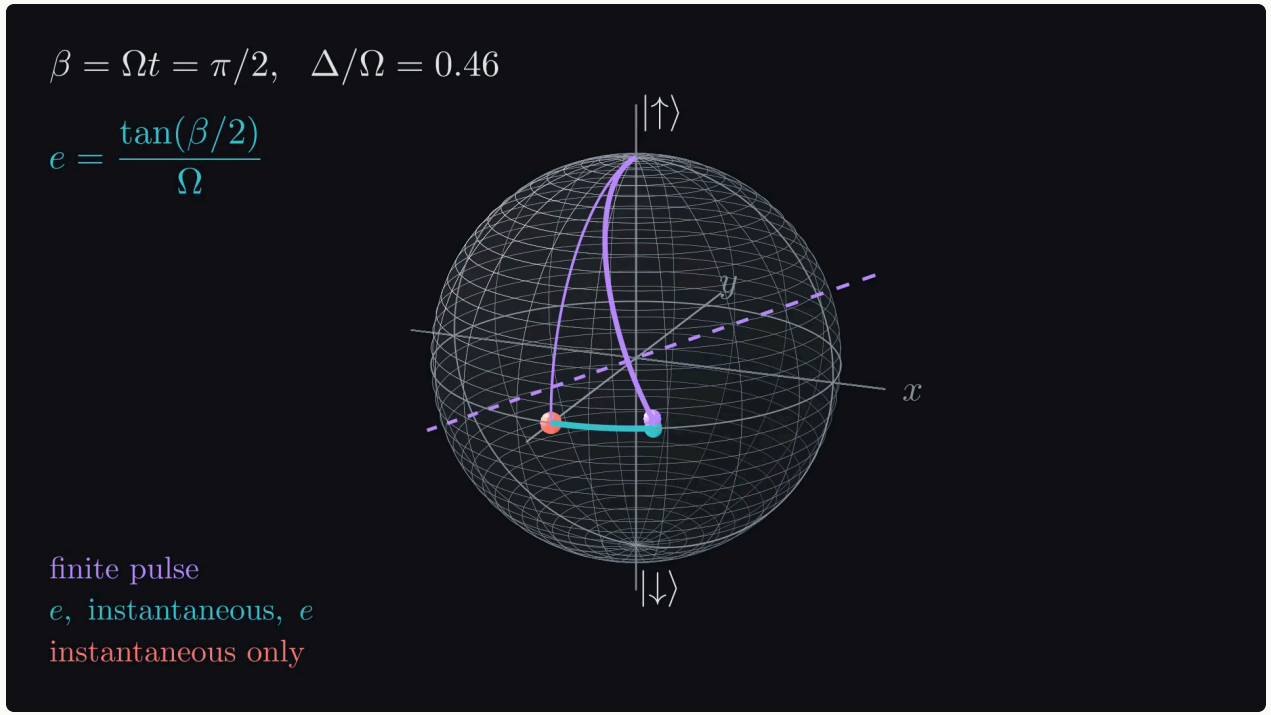


METHOD

Finite pulses

A real pulse depends on the detuning: its axis tilts out of the equator.

$$U = \exp\left[-\frac{i}{2}\,t\left(\Delta\sigma_z + \Omega(\cos\varphi\,\sigma_x + \sin\varphi\,\sigma_y)\right)\right], \qquad \text{axis tilted by } \arctan(\Delta/\Omega)$$



A finite pulse acts like an instantaneous pulse between two short free evolutions:

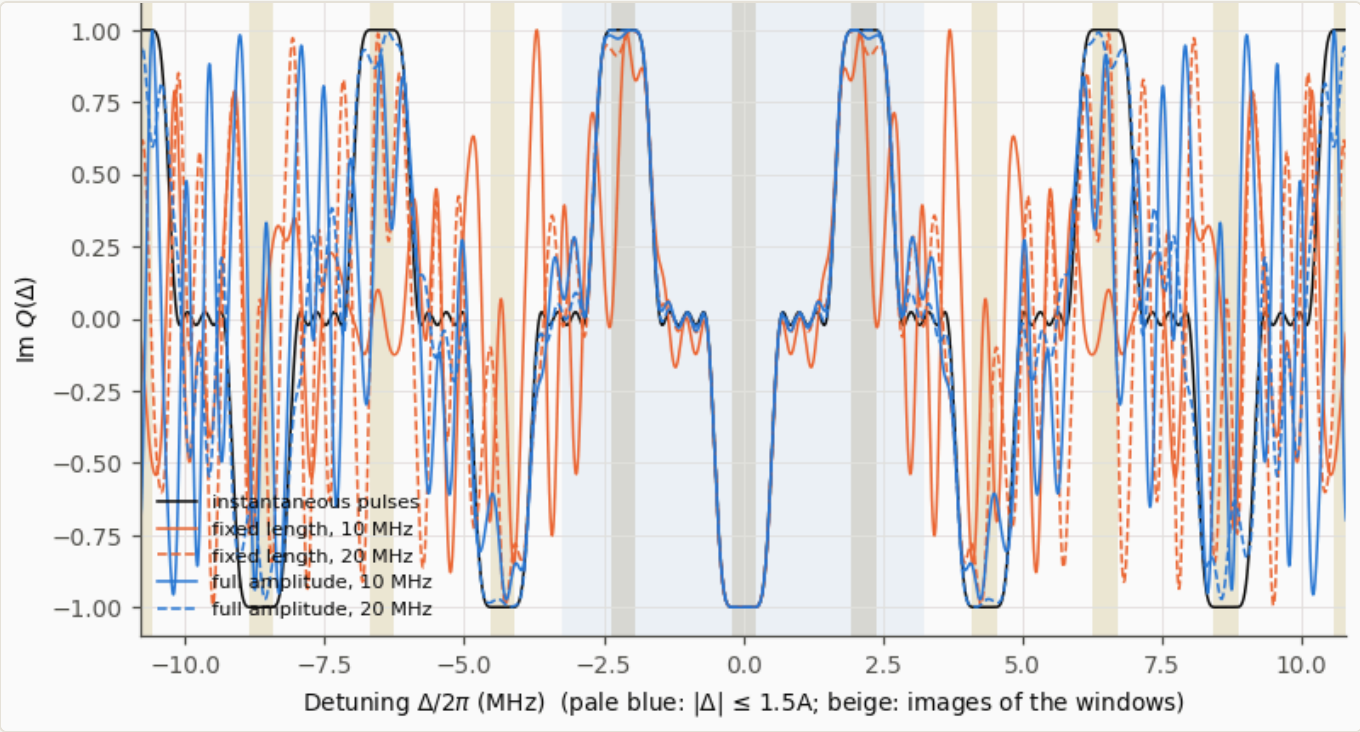
$$e_k = \frac{\tan(\beta_k/2)}{\Omega_k}, \qquad g_j = \frac{2\pi}{A} - e_j - e_{j+1}$$

- Inside $1.5A$, $|\Delta|/\Omega_{\max}$ reaches 0.46 at $10\,\mathrm{MHz}$.
- $\frac{\pi}{2} - \left(\frac{2\pi}{A} - \frac{2}{\Omega_{\max}}\right) - \frac{\pi}{2}$ is exact at the three resonances to 1.4×10^{-6} (drive of Mamin et al.).

RESULTS

Finite pulses, not retrained

Played with real pulses, the $N = 14$ design loses its accuracy.



$N = 14$, not retrained	E
instantaneous	5.1×10^{-4}
full amplitude, 20 MHz	8.0×10^{-3}
full amplitude, 10 MHz	7.9×10^{-2}
fixed length, 20 MHz	0.17
fixed length, 10 MHz	1.0

- The response is no longer periodic.
- Images far away are suppressed.
- Inside $1.5A$ the pulses distort.

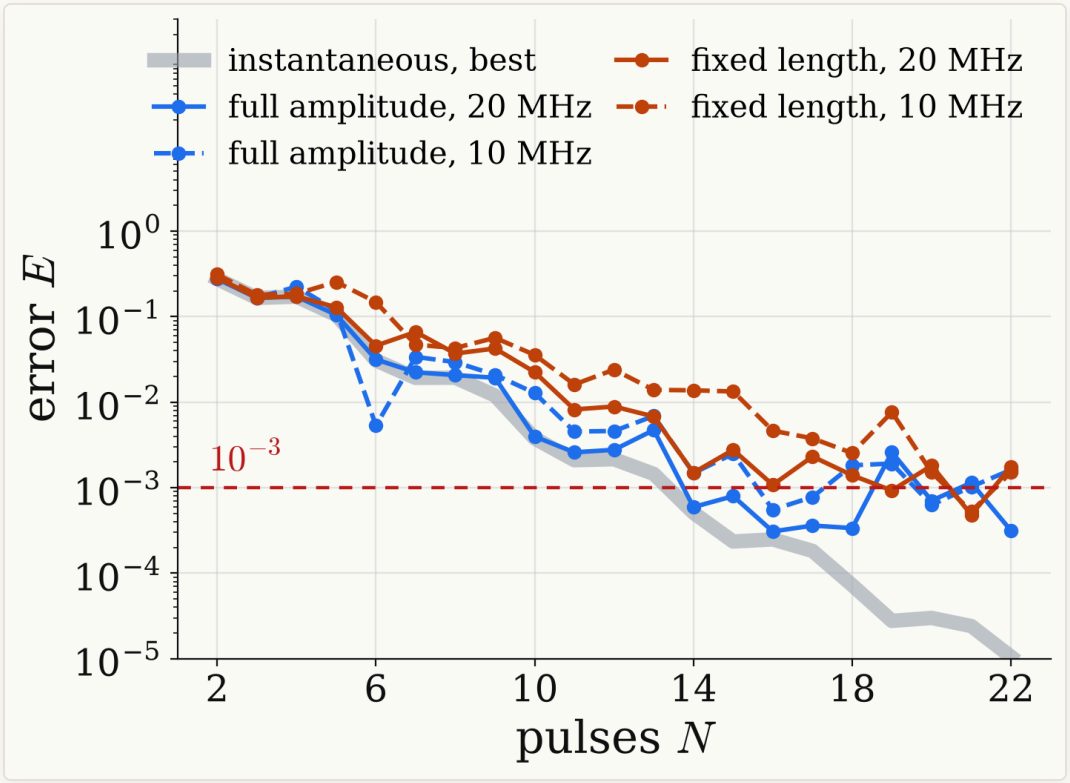
RESULTS

Retraining

Retrained at full amplitude, real pulses cost 0 to 2 extra pulses. Smallest N with $E \leq 10^{-3}$ (total duration):

pulses	$\Omega/2\pi$	symmetric	arbitrary
instantaneous		14 (6.0 μs)	14
full amplitude	20 MHz	14 (5.9 μs)	14 (5.9 μs)
full amplitude	10 MHz	16 (6.7 μs)	15 (6.4 μs)
fixed length	20 MHz	19 (7.8 μs)	> 18
fixed length	10 MHz	21 (9.6 μs)	> 18

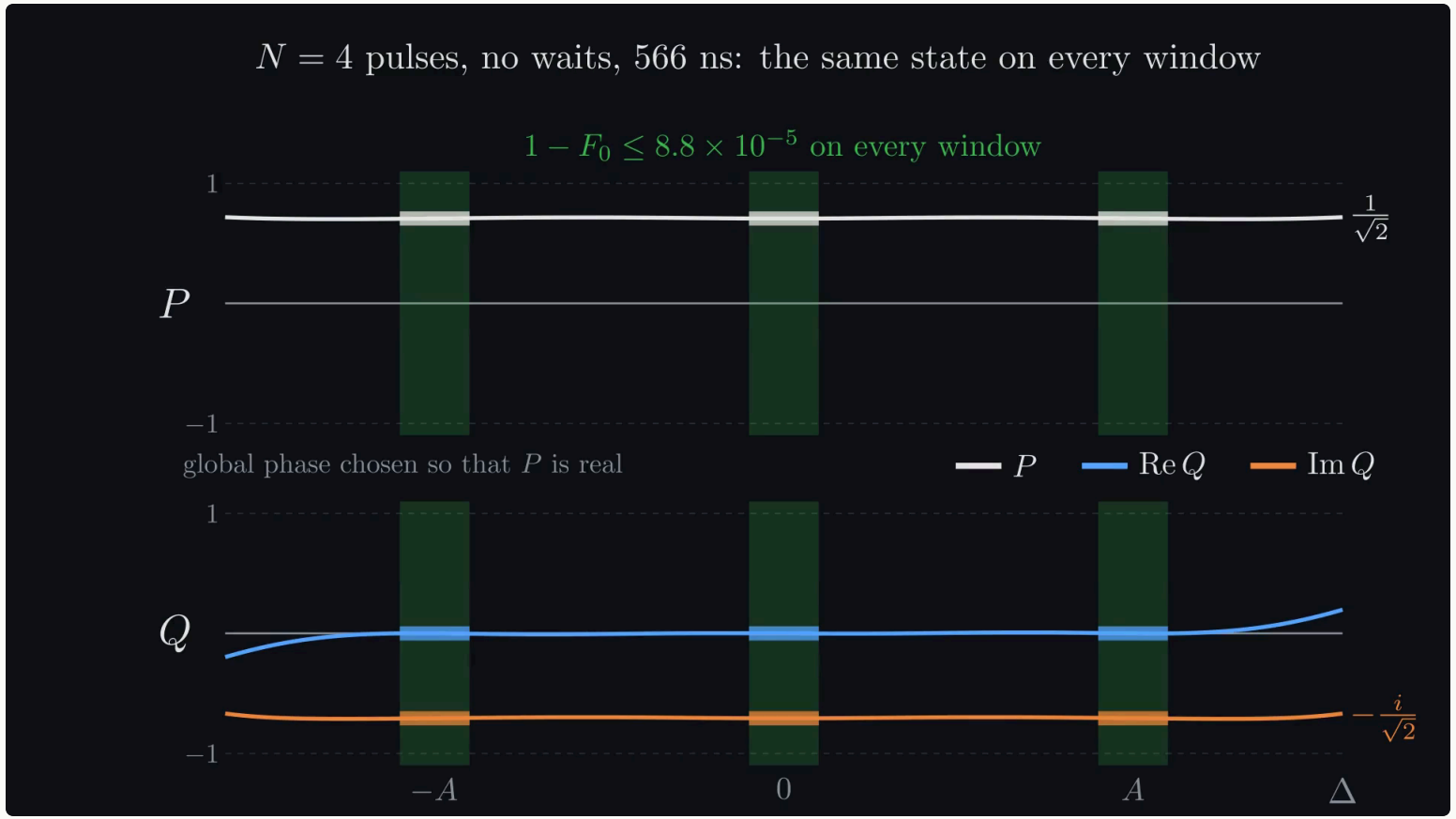
- At fixed length a small angle is a weak pulse.
- Extra pulses per period: 5.2×10^{-4} in $5.9\,\mu$s at $10\,\mathrm{MHz}$.



STRATEGY

State preparation

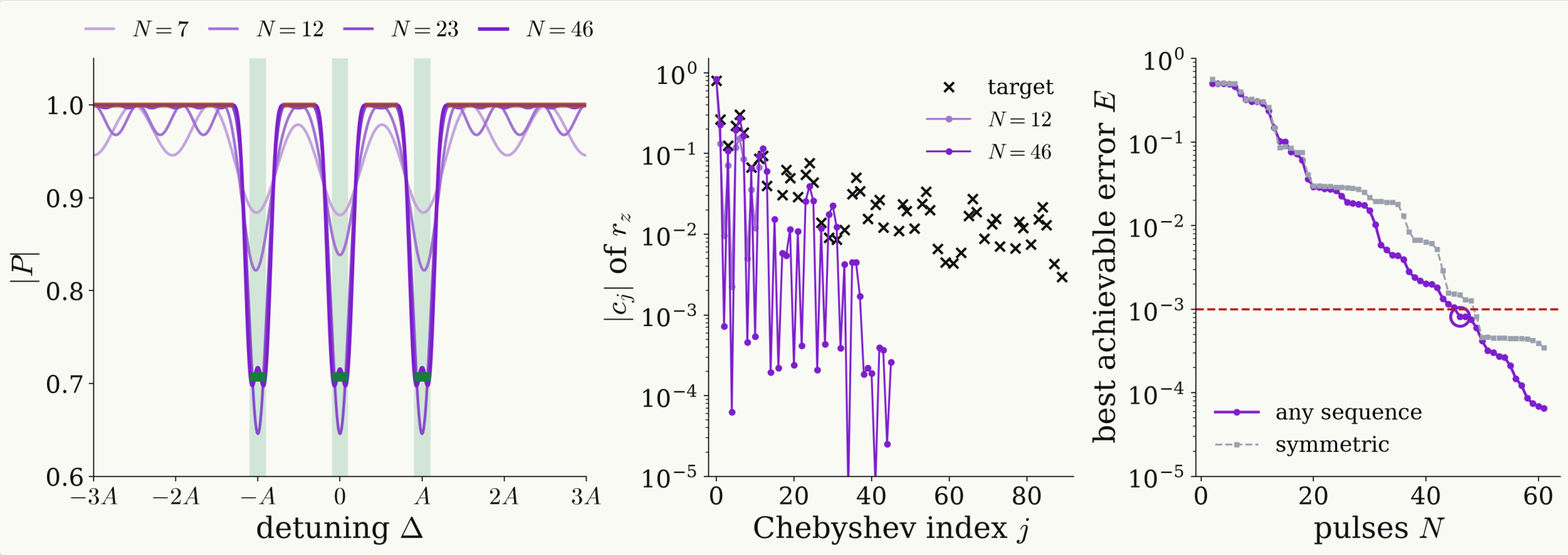
The clip shows the target for preparing $\frac{1}{\sqrt{2}}(|+1\rangle + |-1\rangle)$, a $\pi/2$ pulse about x on all three windows, and then P and Q of one rectangular $\pi/2$ pulse and of a robust four-pulse sequence without waits.



RESULTS

State preparation: Chebyshev expansion

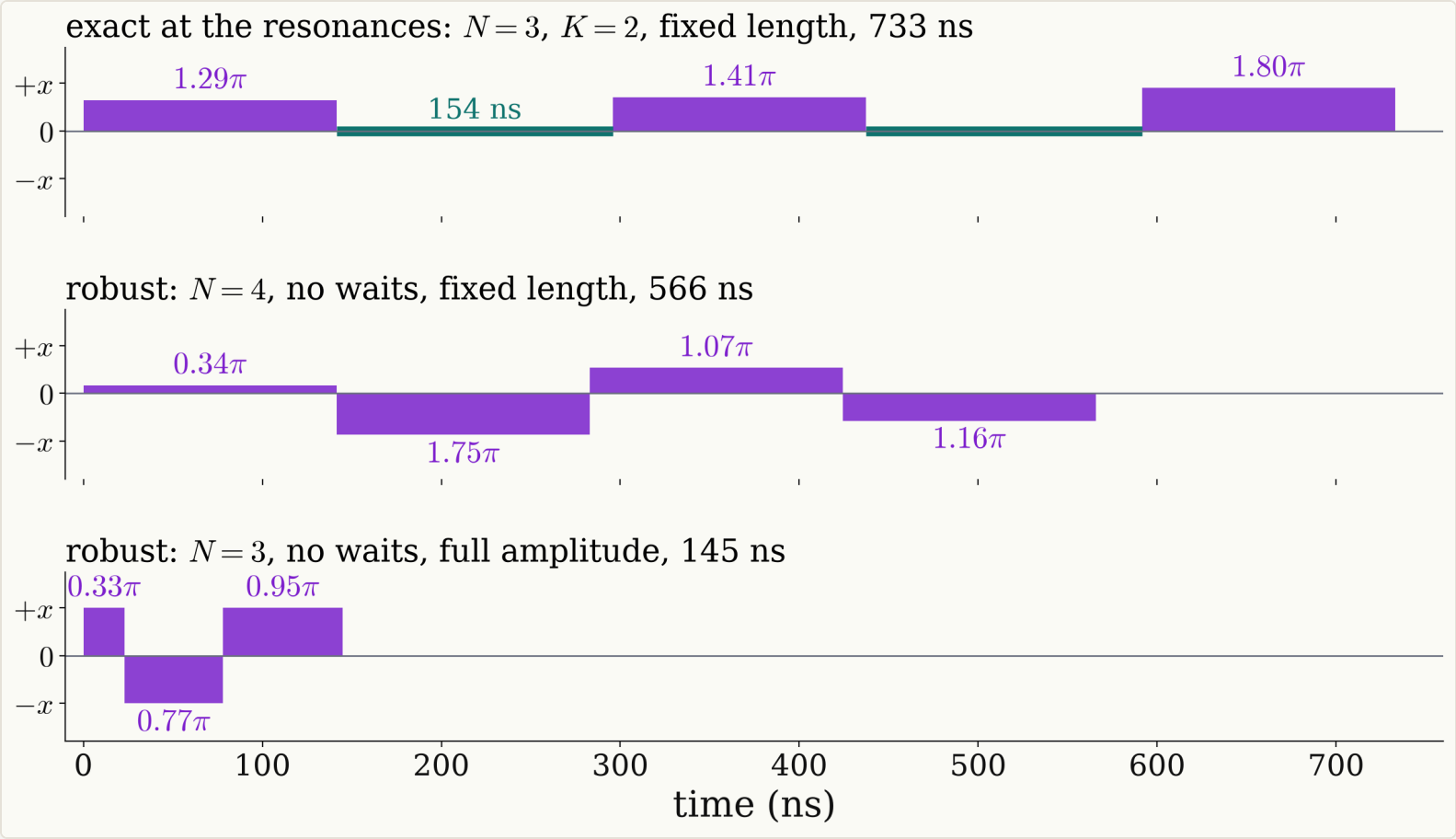
The robust preparation target is a constant, a single Chebyshev coefficient, so it needs no waits; only if $|0\rangle$ must also stay untouched elsewhere in $|\Delta| \leq 3A$ does it become a step, whose coefficients decay as $1/j$ and whose best approximation needs $N = 46$ pulses.



RESULTS

State preparation: pulse sequence

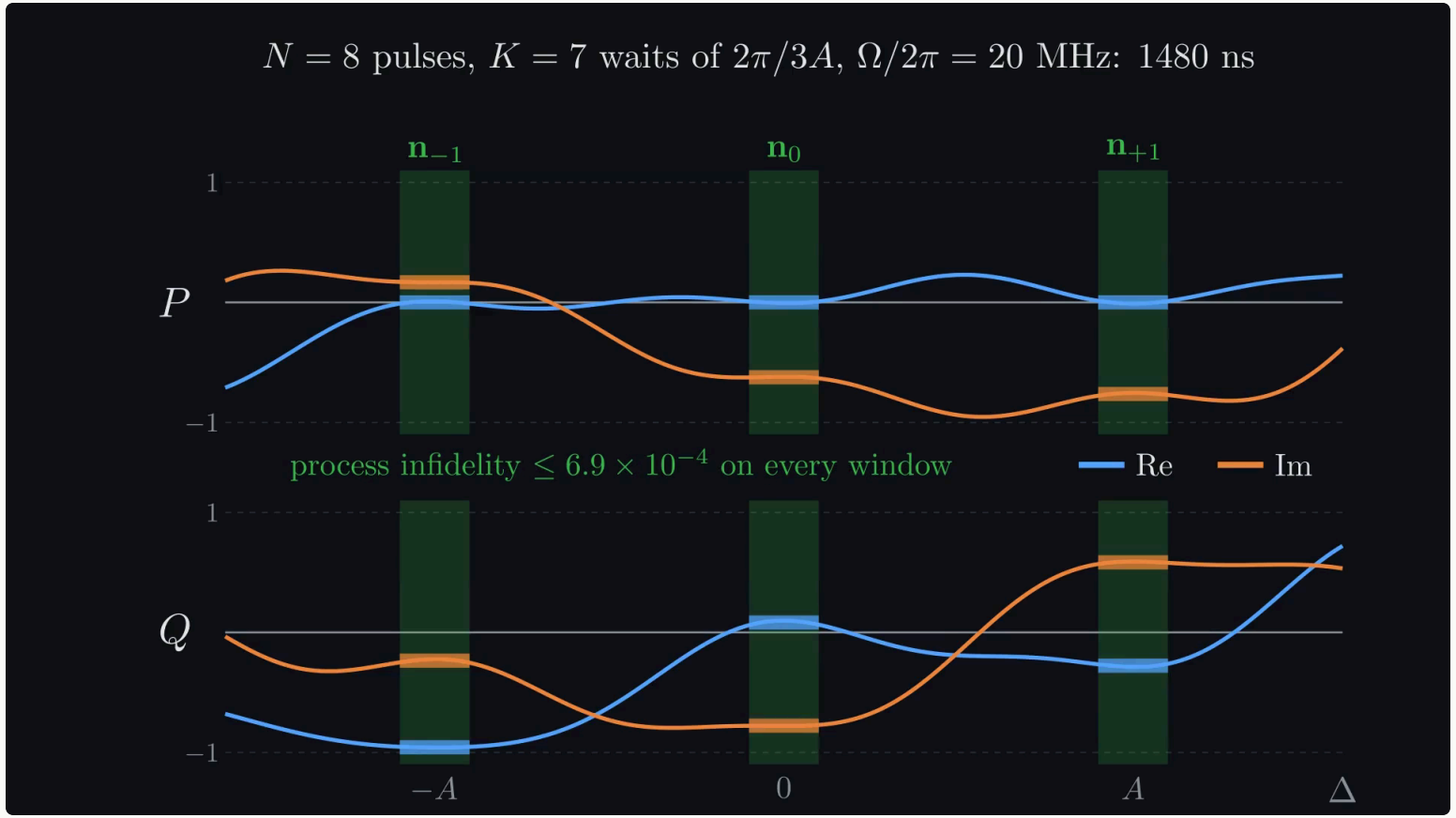
The three preparations at $\Omega/2\pi = 10$ MHz are drawn as pulse trains, with the bar height the amplitude relative to Ω_{DQ}, the sign the axis $\pm x$, the width the duration and the number the rotation angle.



STRATEGY

Universal NOT

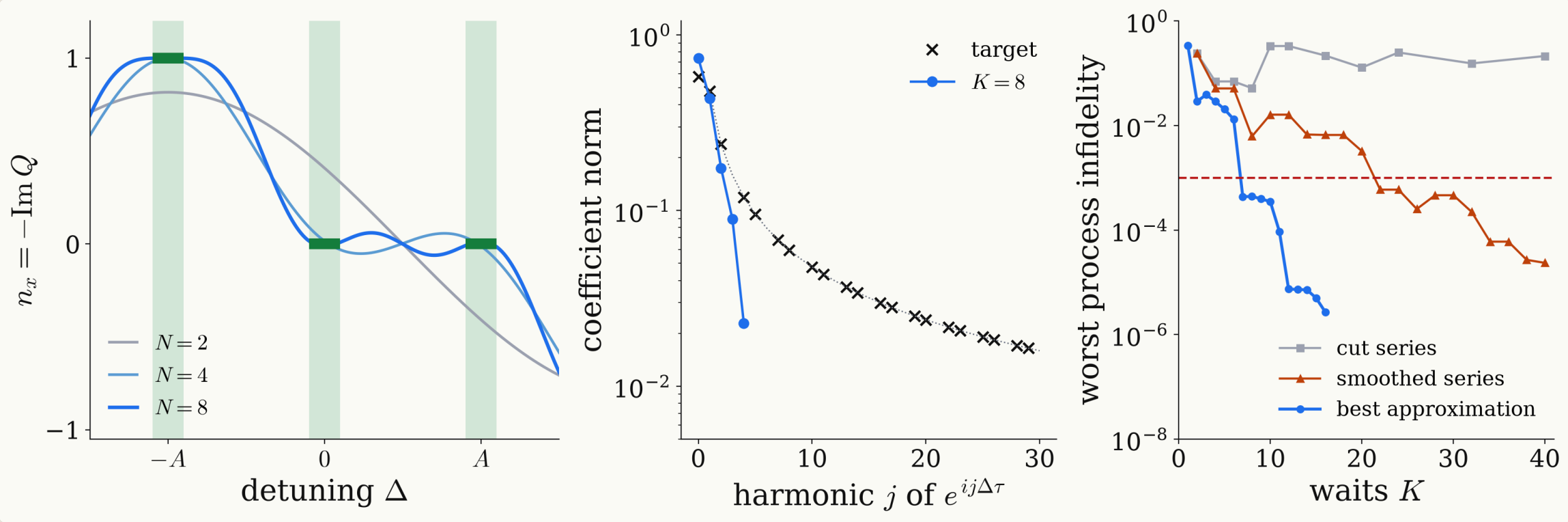
With an unpolarised ^{14}N the electron undergoes $\frac{1}{3}\sum_m U_m \rho U_m^\dagger$, the universal NOT $\mathbf{r} \mapsto -\mathbf{r}/3$ if the U_m are π rotations about orthogonal axes; the clip shows this target and a robust eight-pulse response.



RESULTS

Universal NOT: Laurent expansion

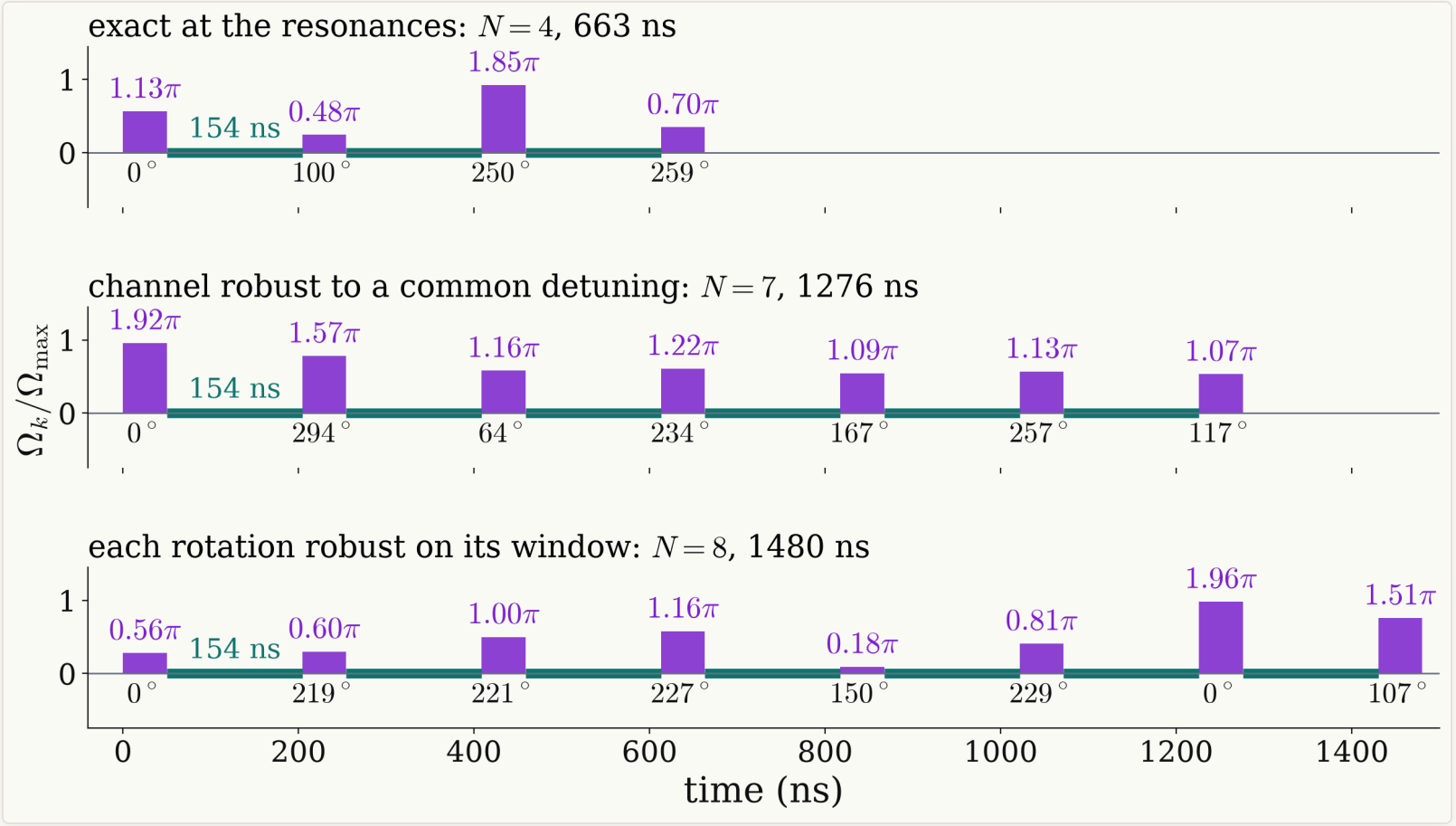
The Laurent coefficients of the universal-NOT target decay only as $3/2\pi|j|$, so the cut series never meets the tolerance; the best approximation meets it with $K = 7$ waits ($N = 8$ pulses).



RESULTS

Universal NOT: pulse sequence

Three universal-NOT sequences at $\Omega/2\pi = 20$ MHz are drawn as pulse trains: bar height the amplitude, rotation angle above, phase below, waits of $2\pi/3A$ in teal.



Take-home

$$\rho_e \mapsto \frac{1}{3}\sum_m U_m\,\rho_e\,U_m^\dagger, \qquad Q = -i\,r(x), \qquad r(x) = \sum_{k\equiv K\,(2)} c_k\,T_k(x)$$

1. The DQ problem we handle is mixed and detuned: the unpolarised ^{14}N gives three detunings $\Delta_m = mA$, and the electron sees their average.
2. QSP tunes a Laurent or Chebyshev polynomial. The best polynomial approximation is the upper bound on what N pulses can do, so setting the target function is what matters.
3. Setting τ is not free: it places the windows on the circle, and no image of a window may land in the stopband.
4. Real pulses break the periodicity and must be retrained; at full amplitude that costs 0 to 2 extra pulses.

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Thank you

Questions & discussion

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B

Backup

The Pauli form, designs exact at the three resonances, a wider stopband, real-pulse training, degree scaling

BACKUP

Pauli form

The Pauli form of a $\pm x$ sequence holds in every basis.

$$u = A(x)\,\mathbb{I} - iB(x)\,X - i\sin\theta\,\big[C(x)\,Z + D(x)\,Y\big], \qquad A^2 + B^2 + (1-x^2)(C^2+D^2) = 1$$

A, B real, degree $\le K$, parity K; C, D real, degree $\le K-1$, parity $K-1$ (K waits).

	Z basis	**$\pm$ basis**
stay	$P = A - i\sin\theta\, C$	$P_\pm = A - iB$
flip	$Q = -iB + \sin\theta\, D$	$\langle -\lvert u\rvert + \rangle = -i\sin\theta\, Q_\pm,\ \ Q_\pm = C - iD$
unitarity	$\lvert P\rvert^2 + \lvert Q\rvert^2 = 1$	$\lvert P_\pm\rvert^2 + (1-x^2)\lvert Q_\pm\rvert^2 = 1$

- A wait maps $(A, B, C, D) \to \big(xA - (1-x^2)C,\ xB - (1-x^2)D,\ A + xC,\ B + xD\big)$.
- $u(-\Delta) = X\,u(\Delta)\,X$, so $P(-\Delta) = P(\Delta)^*$ and $Q(-\Delta) = -Q(\Delta)^*$.
- A palindromic sequence has $u^T = u$, so $D = 0$ and $Q = -iB(x)$.

BACKUP

Example ($K = 2$)

Multiplying out $u = W\,R_x(\phi)\,W$ in both bases checks unitarity; $c = \cos\frac{\phi}{2}$, $s = \sin\frac{\phi}{2}$.

Z basis.

$$u = \begin{pmatrix} z^{-1} & 0 \\ 0 & z \end{pmatrix} \begin{pmatrix} c & -is \\ -is & c \end{pmatrix} \begin{pmatrix} z^{-1} & 0 \\ 0 & z \end{pmatrix} = \begin{pmatrix} c\,z^{-2} & -is \\ -is & c\,z^{2} \end{pmatrix}, \qquad |P|^2 + |Q|^2 = c^2 + s^2 = 1$$

$\pm$ basis. $R_x(\phi) = \mathrm{diag}(e^{-i\phi/2}, e^{i\phi/2})$, $W = \begin{pmatrix} x & -i\sin\theta \\ -i\sin\theta & x \end{pmatrix}$:

$$P_\pm = \cos^2\theta\, e^{-i\phi/2} + (-i\sin\theta)^2 e^{i\phi/2} = c\,T_2(x) - is, \qquad Q_\pm = c\,U_1(x)$$

$$|P_\pm|^2 + (1-x^2)|Q_\pm|^2 = c^2\left[T_2^2 + (1-x^2)U_1^2\right] + s^2 = c^2 + s^2 = 1$$

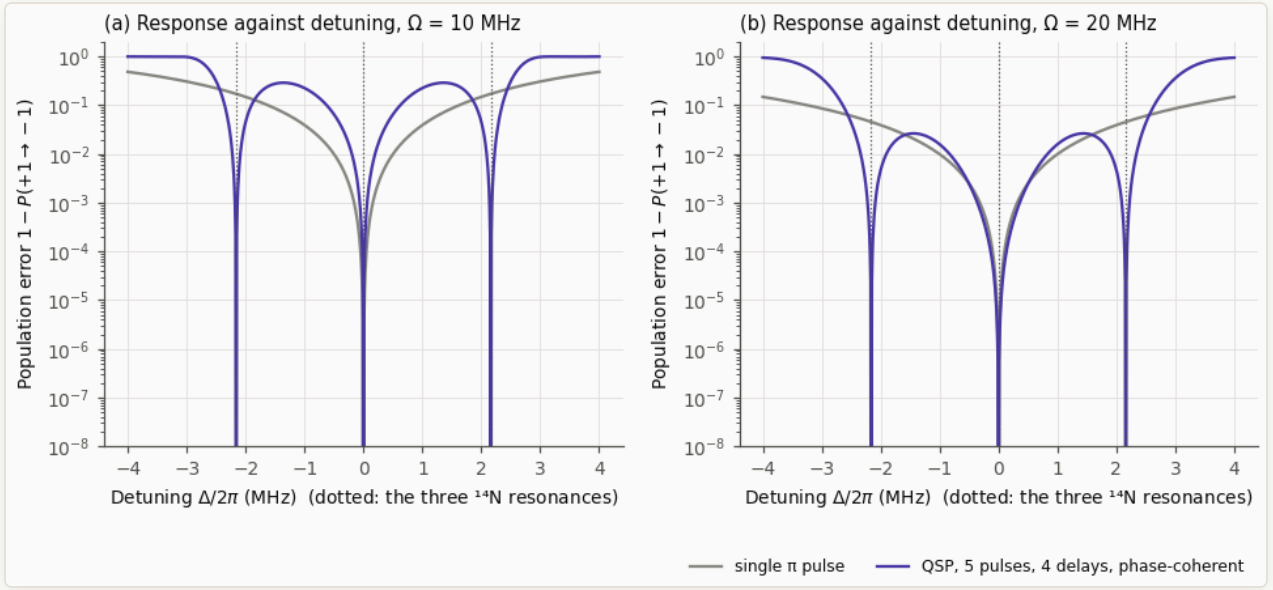
- The bracket is $T_n^2 + (1-x^2)U_{n-1}^2 = \cos^2 n\theta + \sin^2 n\theta = 1$.
- $q = |Q|^2 = s^2$ at every Δ.

BACKUP

Exact designs

Earlier designs are exact at the three resonances only.

$$G_1\,W\,G_2\,W\,G_3\,W\,G_2\,W\,G_1, \qquad \tau = \frac{2\pi}{3A}$$



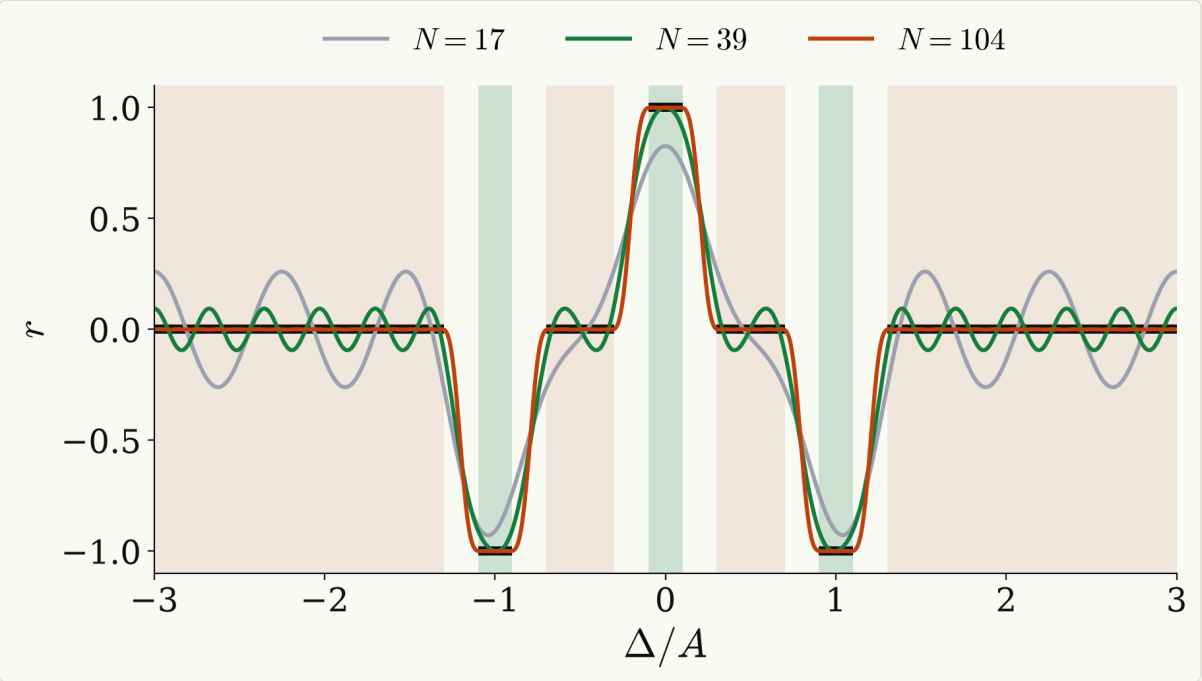
drive	fixed duration	full amplitude
Mamin et al.	777 ns	646 ns
20 MHz	971 ns	683 ns
10 MHz	1324 ns	747 ns

- Five palindromic pulses, phase-coherent.
- Nine-level model, one π pulse to design:
  $0.11 \to 3.1 \times 10^{-4}$ (10 MHz),
  $3.0 \times 10^{-2} \to 8.3 \times 10^{-4}$ (20 MHz).

BACKUP

Wide stopband

A stopband over $|\Delta| \le 3A$ needs 104 pulses for $E = 10^{-3}$.



$$\tau = \frac{2\pi}{6A}$$

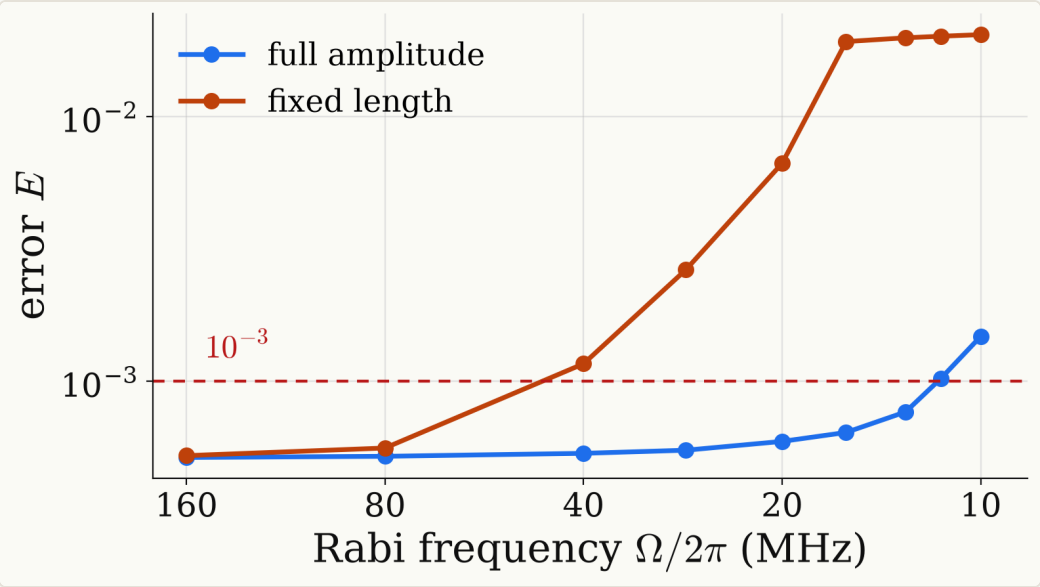
E	N	$K\tau$
0.1	36	2.7 μs
10^{-2}	68	5.2 μs
10^{-3}	104	7.9 μs

- At $\tau = 2\pi/3A$ the smallest degree is $K = 39$; $K\tau$ stays between 5.9 and $6.8\,\mu$s for every τ.
- Smoothed step series in $w = e^{i\Delta\tau}$: degree 48 instead of 21 at $2\pi/3A$, 109 instead of 41 at $2\pi/6A$.

BACKUP

Continuation in the Rabi frequency

The error grows smoothly as the drive weakens.

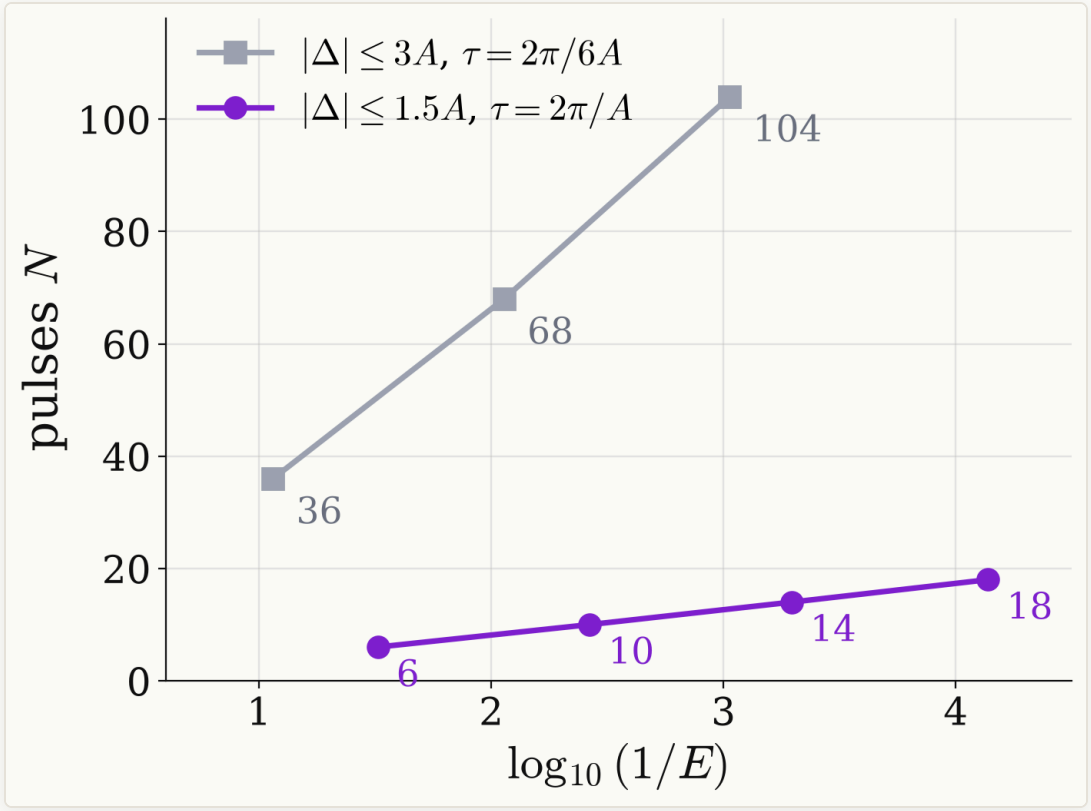


10 MHz, symmetric	$N = 14$ $(2\pi/A)$	$N = 27$ (π/A)	$N = 40$ $(2\pi/3A)$
full amplitude	1.5×10^{-3} (5.8 μs)	5.2×10^{-4} (5.9 μs)	5.2×10^{-4} (5.9 μs)
fixed length	1.4×10^{-2} (5.7 μs)	7.7×10^{-4} (5.8 μs)	2.6×10^{-4} (6.4 μs)

BACKUP

Degree scaling

A step across a gap needs degree $K \propto \ln(1/\varepsilon)/\Delta\theta$.



$$K \simeq c\,\frac{\ln(1/\varepsilon)}{\Delta\theta}$$

- The gap $\Delta\theta$ is the transition band, $0.2A$ in Δ: 0.2π at $\tau = 2\pi/A$, $0.2\pi/6$ at $\tau = 2\pi/6A$.
- Both targets need a number of pulses linear in $\ln(1/E)$.
- E_N falls by 7 to 8 per four pulses.

Low and Chuang (2017); Martyn et al. (2021)